Austo Automobiles Business Report

Customer Purchase Behavior & Marketing Insights

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Executive Summary

This report explores customer purchase behavior at Austo Automobiles using data-driven insights. The primary objective was to analyze key demographic and economic variables influencing automobile purchase decisions, including gender, age, salary, partner income, and vehicle price.

Key findings include:

- Males tend to prefer SUVs more compared to females.

- Salaried individuals are more likely to purchase Sedans.

- Females on average spend more on car purchases.

- Individuals with personal loans are likely to spend higher amounts on automobiles.

Based on these insights, we recommend targeted marketing strategies, such as promoting premium models to buyers aged 40+, bundling Sedans with attractive financing for salaried individuals, and developing SUV campaigns directed at male customers.

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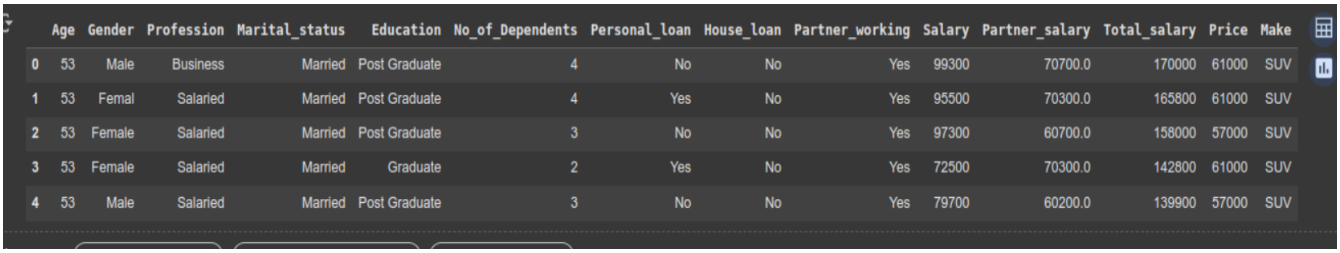
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Introduction

Austo Motor Company, a leading manufacturer of SUVs, Sedans, and Hatchbacks, aims to enhance the effectiveness of its marketing strategy by better understanding customer preferences and purchase behavior. In response to concerns raised during a recent board meeting, the Data Science team was tasked with analyzing customer data to uncover key insights.

This report presents a structured analysis of the dataset, including data cleaning, univariate and bivariate exploration, and targeted answers to specific business questions. The goal is to derive actionable insights that can support data-driven marketing decisions and improve customer engagement.

Data Overview and Treatment



	Age	Gender	Profession	Marital_status	Education	No_of_Dependents	Personal_loan	House_loan	Partner_working	Salary	Partner_salary	Total_salary	Price	Make
0	53	Male	Business	Married	Post Graduate	4	No	No	Yes	99300	70700.0	170000	61000	SUV
1	53	Femal	Salaried	Married	Post Graduate	4	Yes	No	Yes	95500	70300.0	165800	61000	SUV
2	53	Female	Salaried	Married	Post Graduate	3	No	No	Yes	97300	60700.0	158000	57000	SUV
3	53	Female	Salaried	Married	Graduate	2	Yes	No	Yes	72500	70300.0	142800	61000	SUV
4	53	Male	Salaried	Married	Post Graduate	3	No	No	Yes	79700	60200.0	139900	57000	SUV

Fig1. Source: Austo Automobiles Notebook

Noticed typo in Gender, "Femal" so we explored further. All sample rows have "SUV" in Make, hinting at class imbalance we'll verify later.

```
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Data columns (total 14 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
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 1   Gender            1528 non-null   object
 2   Profession        1581 non-null   object
 3   Marital_status    1581 non-null   object
 4   Education         1581 non-null   object
 5   No_of_Dependents  1581 non-null   int64
 6   Personal_loan     1581 non-null   object
 7   House_loan        1581 non-null   object
 8   Partner_working   1581 non-null   object
 9   Salary            1581 non-null   int64
 10  Partner_salary    1475 non-null   float64
 11  Total_salary      1581 non-null   int64
 12  Price             1581 non-null   int64
 13  Make              1581 non-null   object
dtypes: float64(1), int64(5), object(8)
memory usage: 173.1+ KB
```

Fig2. Source: Austo Automobiles Notebook

Confirmed the data types above and reveals non-null counts: All columns except Gender (1528 non-null) and Partner_salary (1475 non-null) are complete. All columns seem to have appropriate data types.

	count
Gender	
Male	1199
Female	327
Femal	1
Femle	1

Fig3. Source: Austo Automobiles Notebook

As we can see in the gender column there is missing data, we dropped all those rows and we can also see typos like "Femal" and "Femle" we will fix that too after that.

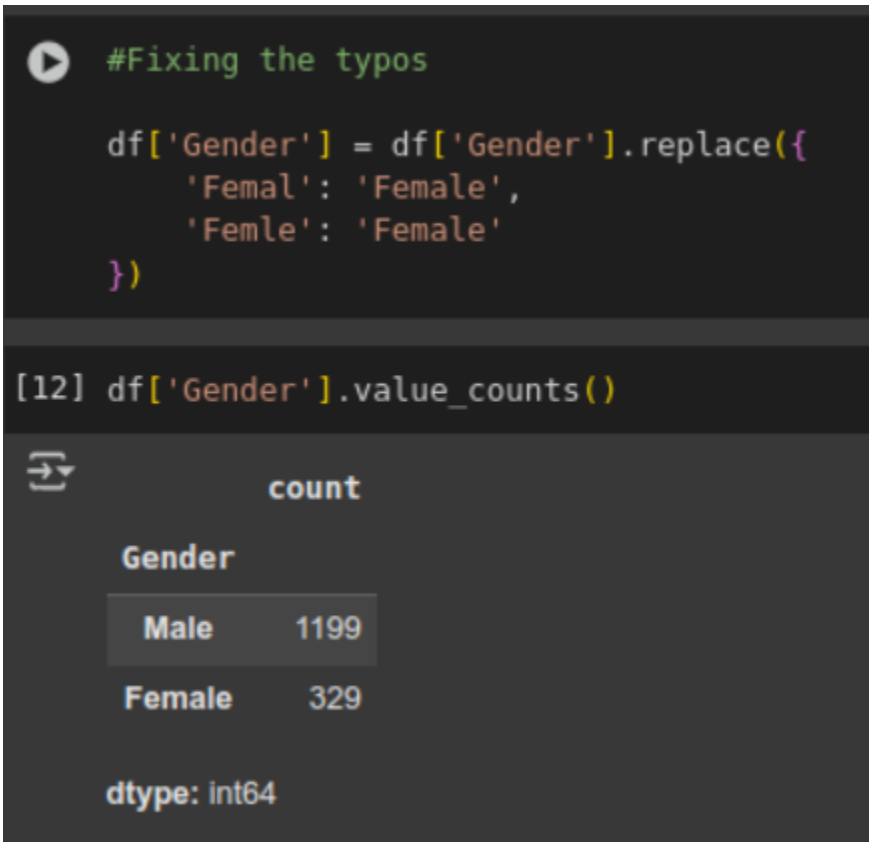


Fig4. Source: Austo Automobiles Notebook

After dropping the null gender rows and fixing typos the gender column is fixed.

As we could see, there were null values in the column "Partner_Salary", but there were no null values in the columns "Partner_working", "Salary", and "Total_salary". To fix "Partner_Salary", we checked: if "Partner_working" was No, then "Partner_Salary" was set to 0; and if "Partner_working" was Yes, we set Partner_Salary = Total_salary - Salary.

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---  ------            --------------  -----  
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 1   Gender            1528 non-null   object 
 2   Profession        1528 non-null   object 
 3   Marital_status    1528 non-null   object 
 4   Education         1528 non-null   object 
 5   No_of_Dependents  1528 non-null   int64  
 6   Personal_loan     1528 non-null   object 
 7   House_loan        1528 non-null   object 
 8   Partner_working   1528 non-null   object 
 9   Salary            1528 non-null   int64  
 10  Partner_salary    1528 non-null   float64
 11  Total_salary      1528 non-null   int64  
 12  Price             1528 non-null   int64  
 13  Make              1528 non-null   object 
dtypes: float64(1), int64(5), object(8)
memory usage: 167.3+ KB
```

Fig5. Source: Austo Automobiles Notebook

After treating the data for Gender and Partner_Salary columns we can see there is no missing data and all data types are correct. No typos are left either.

Data Description

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Age	1528.0	NaN	NaN	NaN	32.013743	8.45167	22.0	25.0	29.0	38.0	54.0
Gender	1528	2	Male	1199	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Profession	1528	2	Salaried	864	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Marital_status	1528	2	Married	1395	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Education	1528	2	Post Graduate	951	NaN	NaN	NaN	NaN	NaN	NaN	NaN
No_of_Dependents	1528.0	NaN	NaN	NaN	2.448953	0.943741	0.0	2.0	2.0	3.0	4.0
Personal_loan	1528	2	Yes	768	NaN	NaN	NaN	NaN	NaN	NaN	NaN
House_loan	1528	2	No	1018	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Partner_working	1528	2	Yes	838	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Salary	1528.0	NaN	NaN	NaN	60453.403141	14682.669181	30000.0	51900.0	59500.0	72000.0	99300.0
Partner_salary	1528.0	NaN	NaN	NaN	19516.361257	19764.60575	0.0	0.0	25300.0	38200.0	80500.0
Total_salary	1528.0	NaN	NaN	NaN	79969.764398	25608.884386	30600.0	60875.0	78150.0	96025.0	171000.0
Price	1528.0	NaN	NaN	NaN	36022.905759	13539.294925	18000.0	25000.0	32000.0	47000.0	70000.0
Make	1528	3	Sedan	657	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Fig6. Source: Austo Automobiles Notebook

Key Observations:

- Age Distribution: The average age is 32, with most individuals between 25 and 38. The range (22–54) shows a fairly young to mid-career working population with no extreme outliers.
- Dependents: Most people have 2 to 3 dependents. No one has more than 4, so the data is realistic and clean.
- Salary: Median salary is ₹59,500 with a modest spread. The upper limit is ₹99,300, indicating no massive income outliers. Distribution is mildly right-skewed due to a few high earners.
- Partner's Salary: The partner salary is zero for a significant number of rows (at least 25% of the dataset), matching with many entries where Partner_working = No. Among working partners, the income varies widely — from low to fairly high (up to ₹80,500).

- Total Salary: This is just the sum of the main and partner salaries. It ranges from ₹30,600 to ₹1,71,000, indicating a few high-income households. Because it's a derived column, you should avoid using it in modeling along with Salary and Partner_salary to prevent redundancy.
- Price of Purchased Product (likely a vehicle): The average price is ₹36,023, median ₹32,000. The data is skewed right — 75% of the population buys under ₹47,000. This means most customers opt for mid-range pricing and only a few go for expensive options (up to ₹70,000).
- Sedan is the most purchased Make.

Univariate Analysis

- *Histograms & box-plots for numeric columns.*
- *For categorical columns we use bar (count) plots or a frequency table.*

1. No_of_Dependents

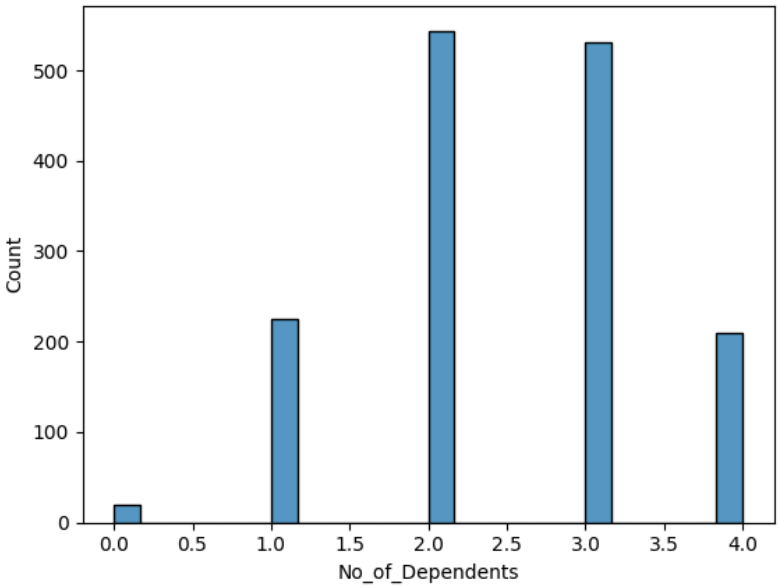


Fig7. Source: Austo Automobiles Notebook

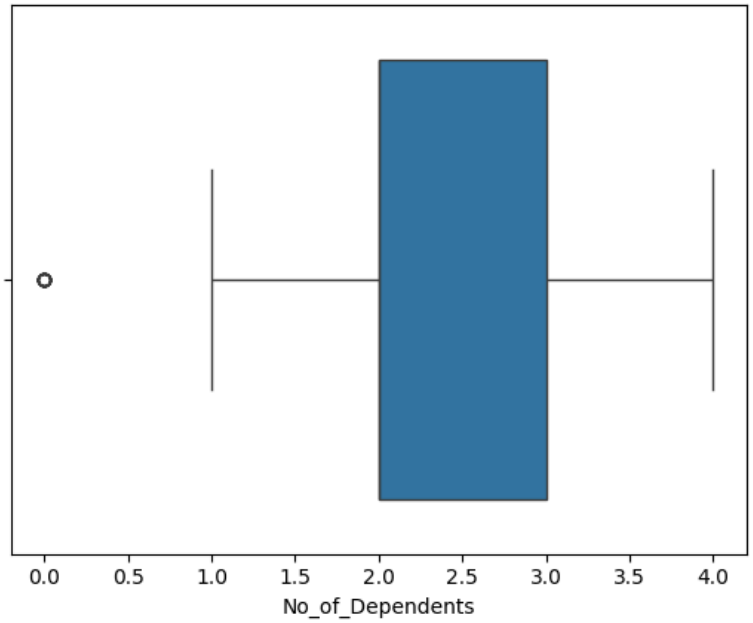


Fig8. Source: Austo Automobiles Notebook

Nearly two-thirds of customers support 2–3 dependents, suggesting mid-sized family units dominate the target market. So, marketing messages framed around 'family needs' will resonate with the vast majority.

Every value lies inside 0-4; keep as-is.

Salary

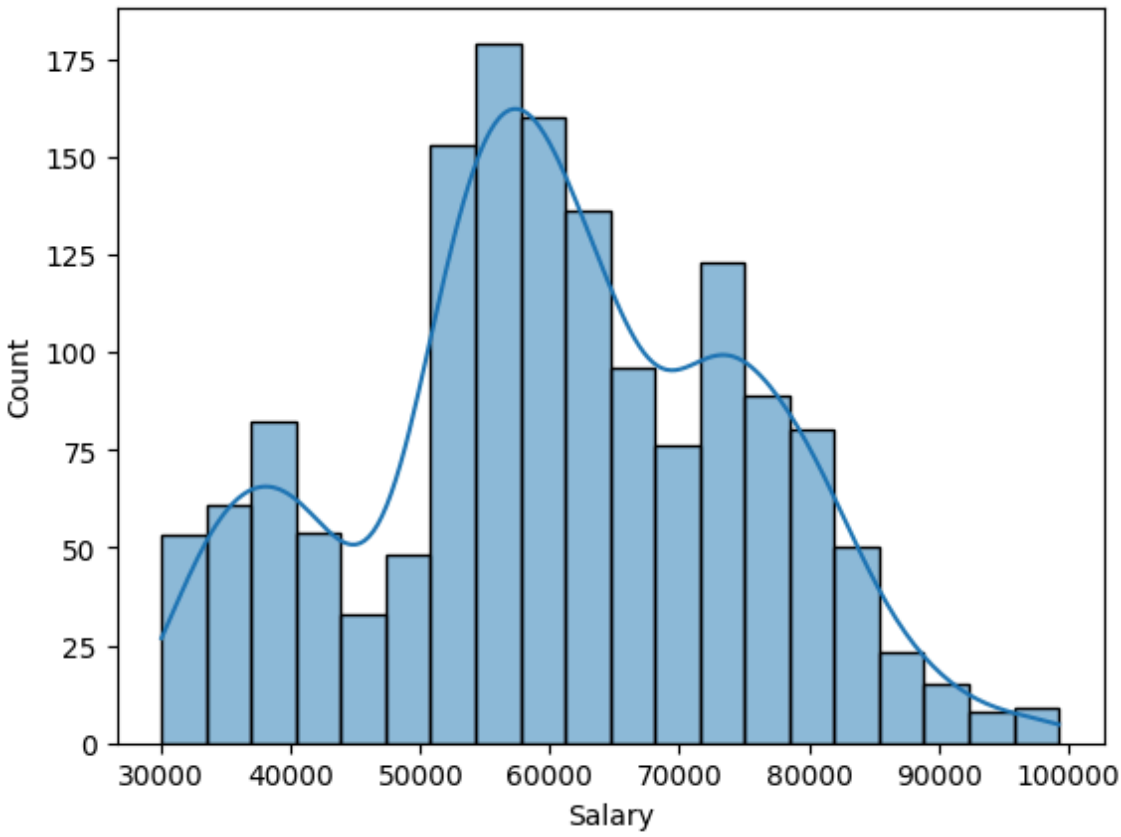


Fig9. Source: Austo Automobiles Notebook

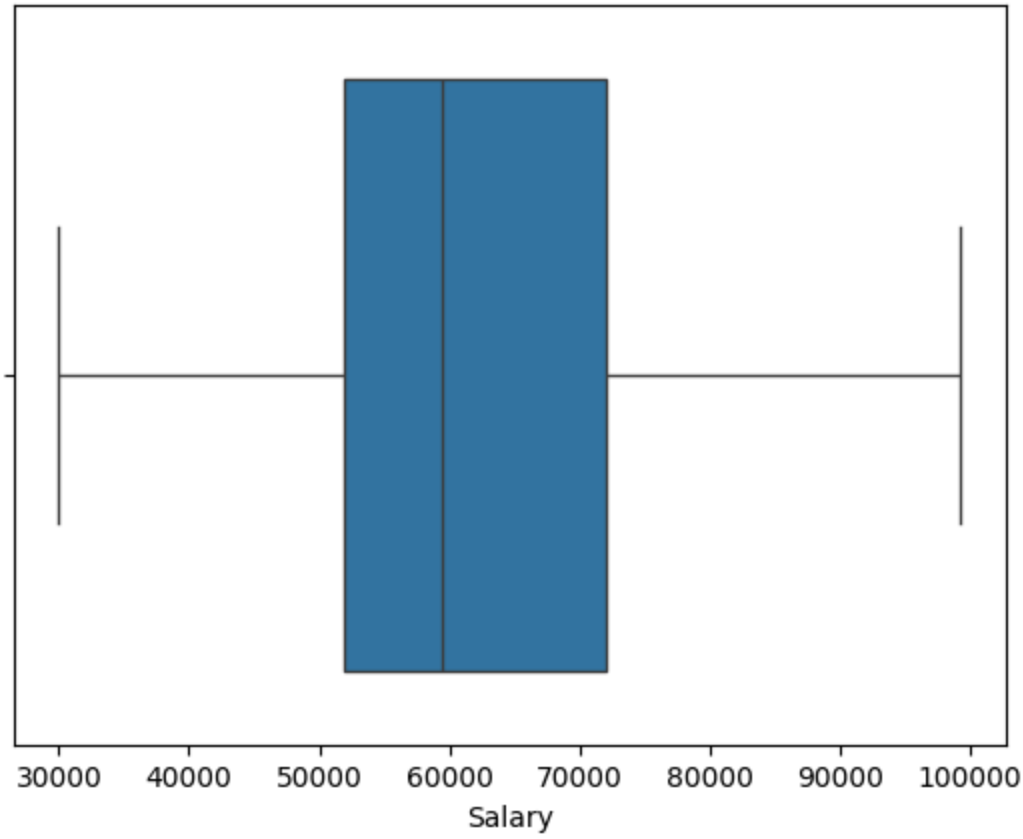
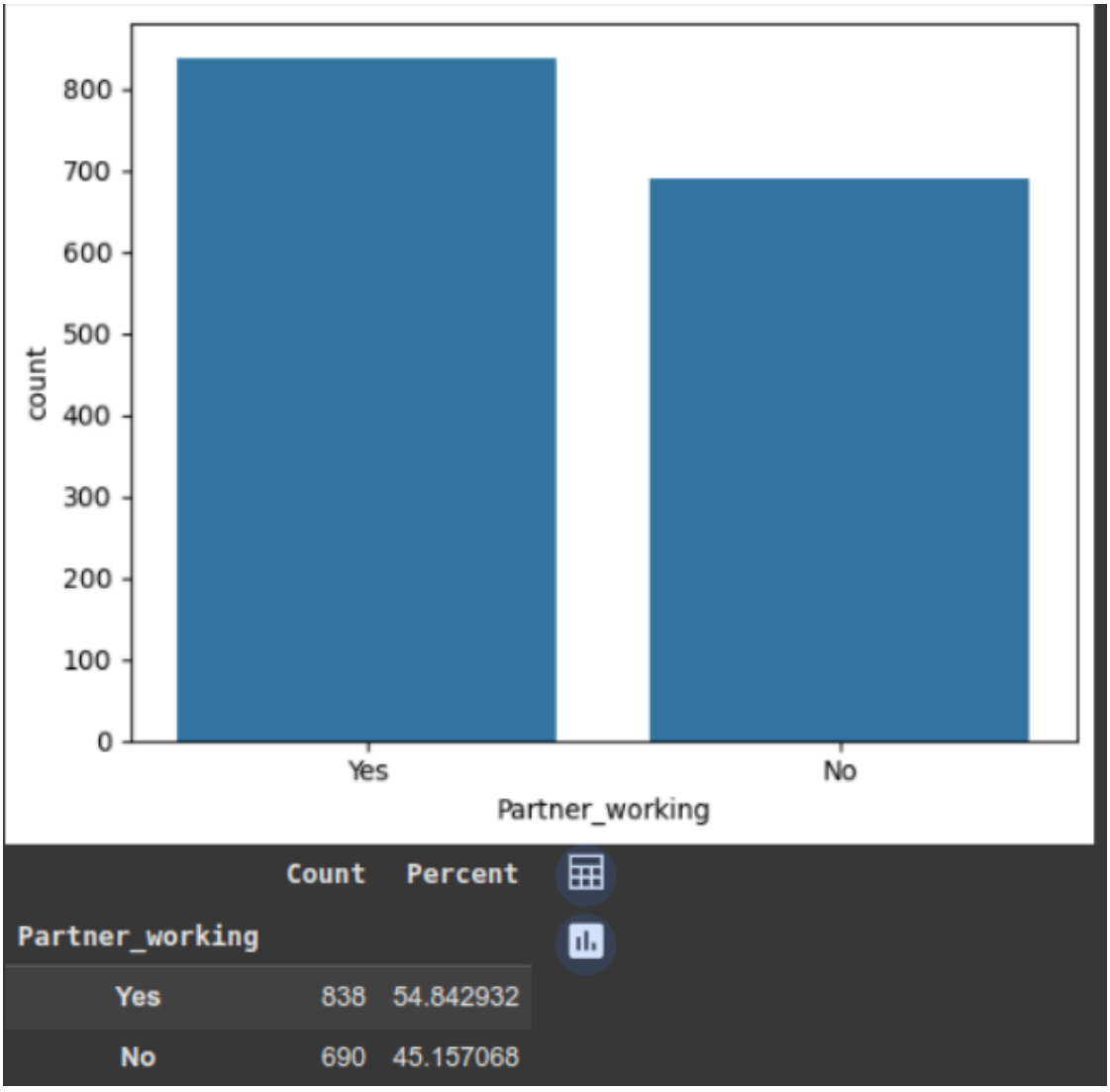


Fig10. Source: Austo Automobiles Notebook

The salary distribution is unimodal and slightly right-skewed, with most customers earning between ₹55,000 and ₹65,000. The median salary is ₹59,500, and the spread of salaries is moderate, ranging from ₹30,000 to ₹99,000. There are no salary outliers, indicating a clean and well-bounded dataset. This implies a predominantly mid-income customer base, useful for setting target price bands for products.

2. Partner_working



Partner_working	Count	Percent
Yes	838	54.842932
No	690	45.157068

Fig11. Source: Austo Automobiles Notebook

A slim majority (≈ 55 %) of customers have a working partner. This aligns with the earlier Partner _salary distribution, where ≈ 45 % of partners earned no income. We'll revisit this split when analyzing how partner income influences vehicle buying trends.

3. Partner_salary

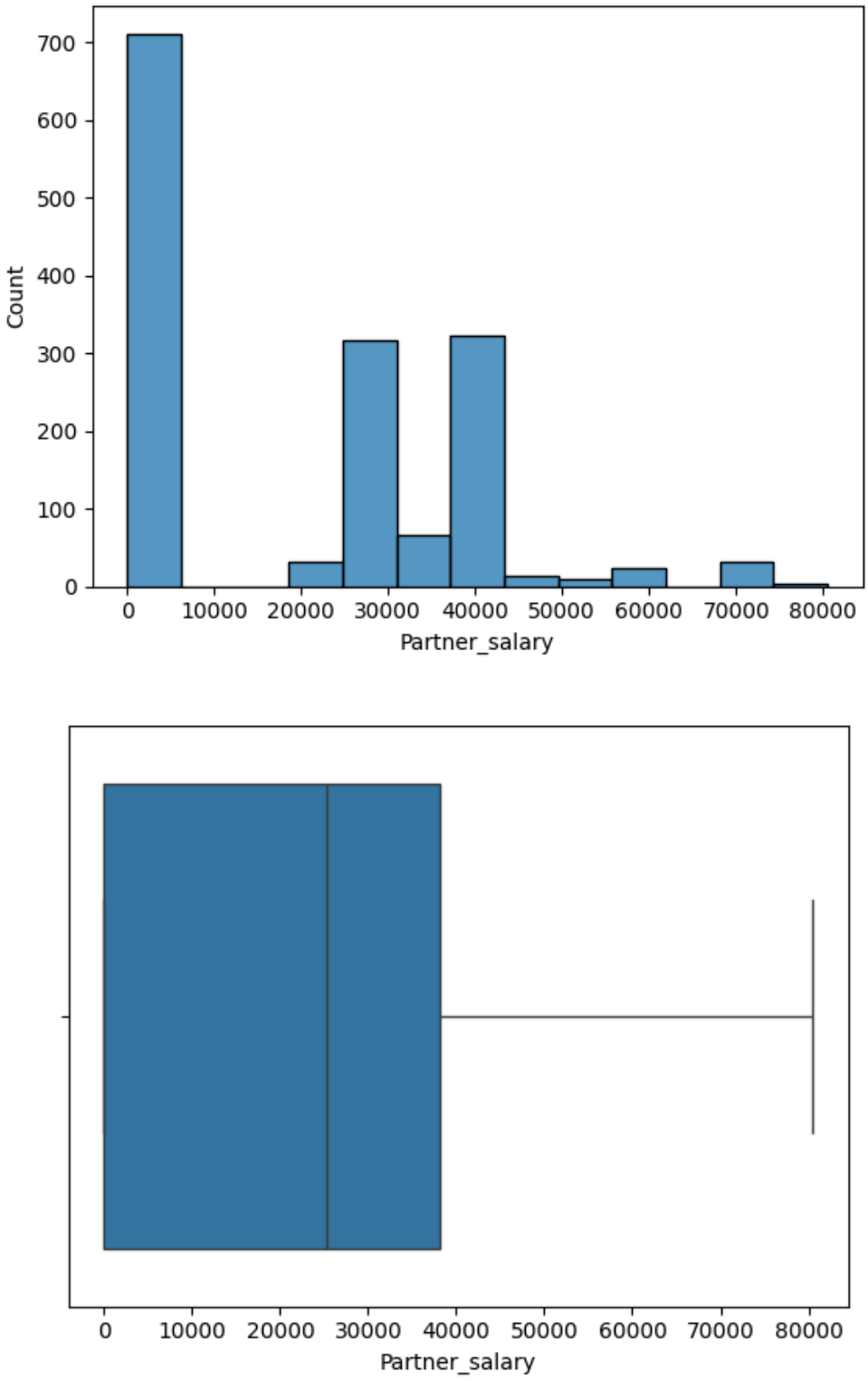


Fig12&13. Source: Austo Automobiles Notebook

The distribution of Partner_salary is heavily right-skewed, with nearly ≈ 45 % of entries showing zero income — consistent with the Partner_working column. Among working partners, salary clusters around ₹30,000–₹40,000, with a few earning above ₹60,000. This split structure suggests the need to analyze this variable conditionally (i.e., split by working status) to avoid misleading central tendencies. No outliers were found, and the data appears valid.

4. **Total_salary**

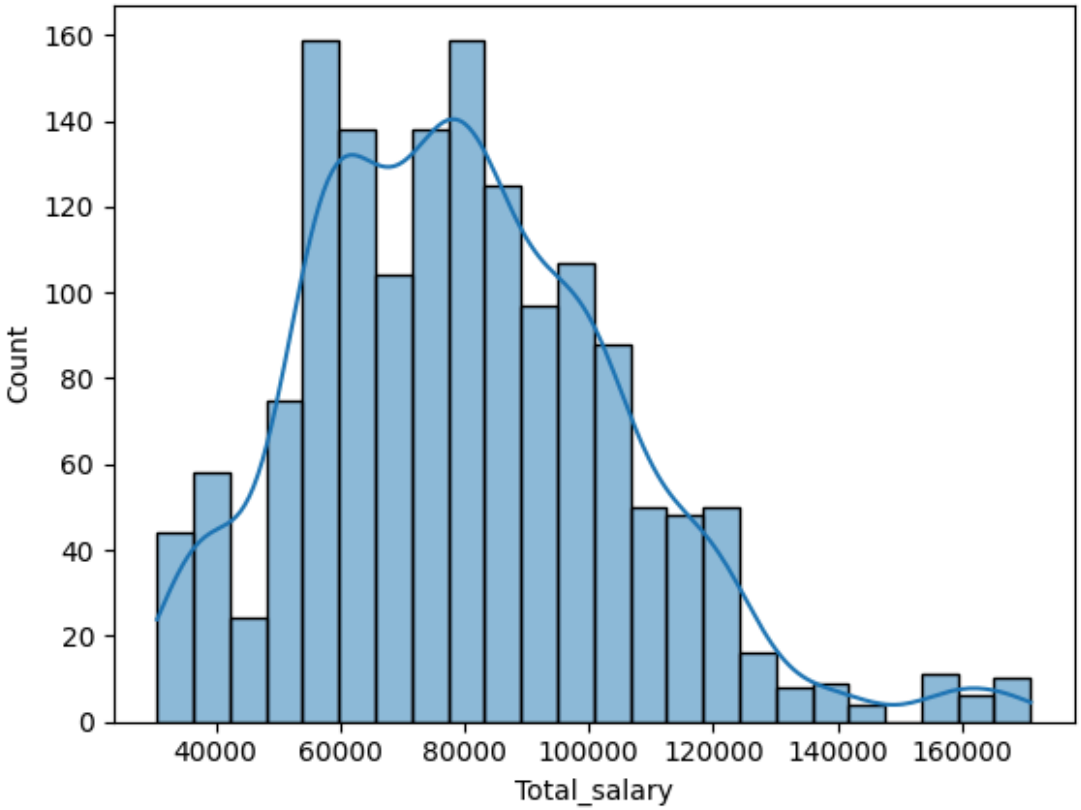


Fig14. Source: Austo Automobiles Notebook

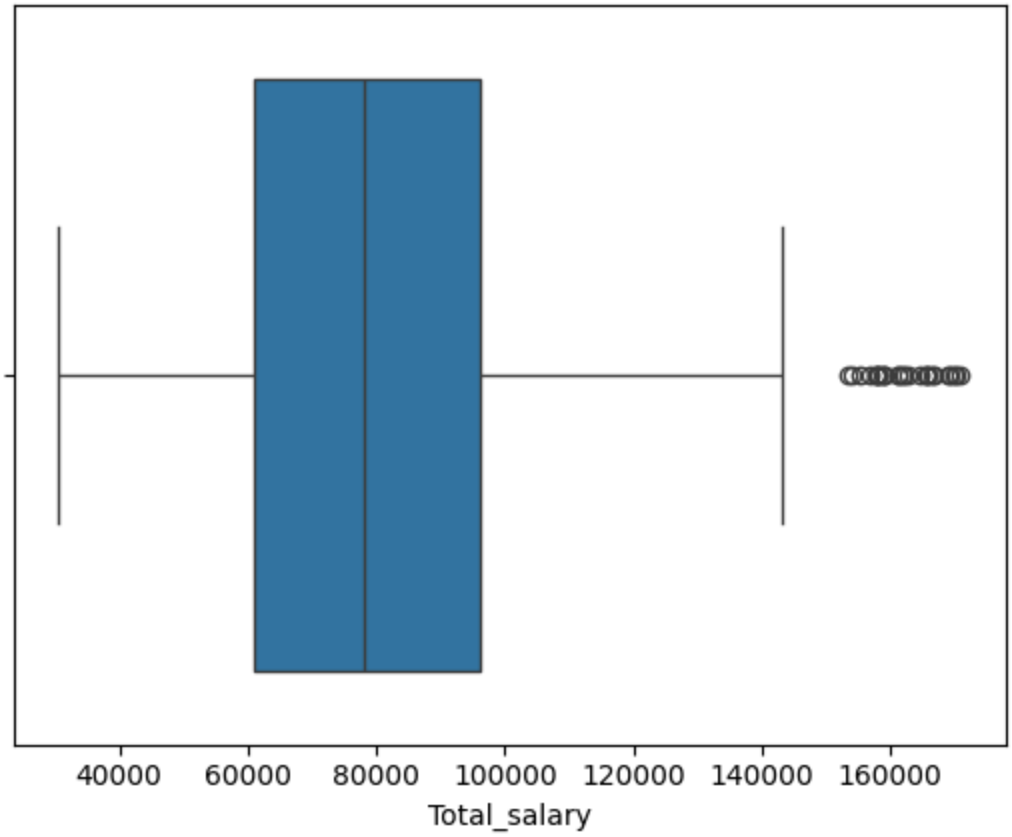
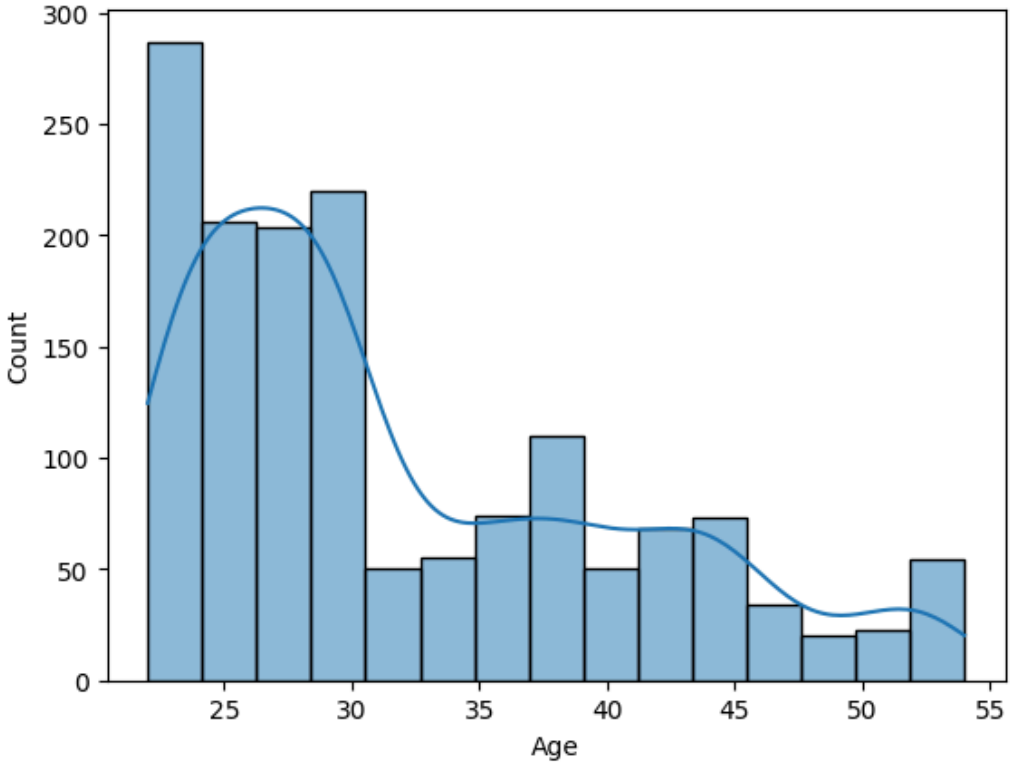


Fig15. Source: Austo Automobiles Notebook

Total_salary is right-skewed, with the bulk of household incomes between ₹60k and ₹100k, and a small affluent tail extending up to ₹171k. Some high-income outliers were observed, which we investigated further in the notebook. These outliers were found to be either from business income sources, two-income households, or both so, these were very plausible outliers so no need for treatment.

5. **Age**



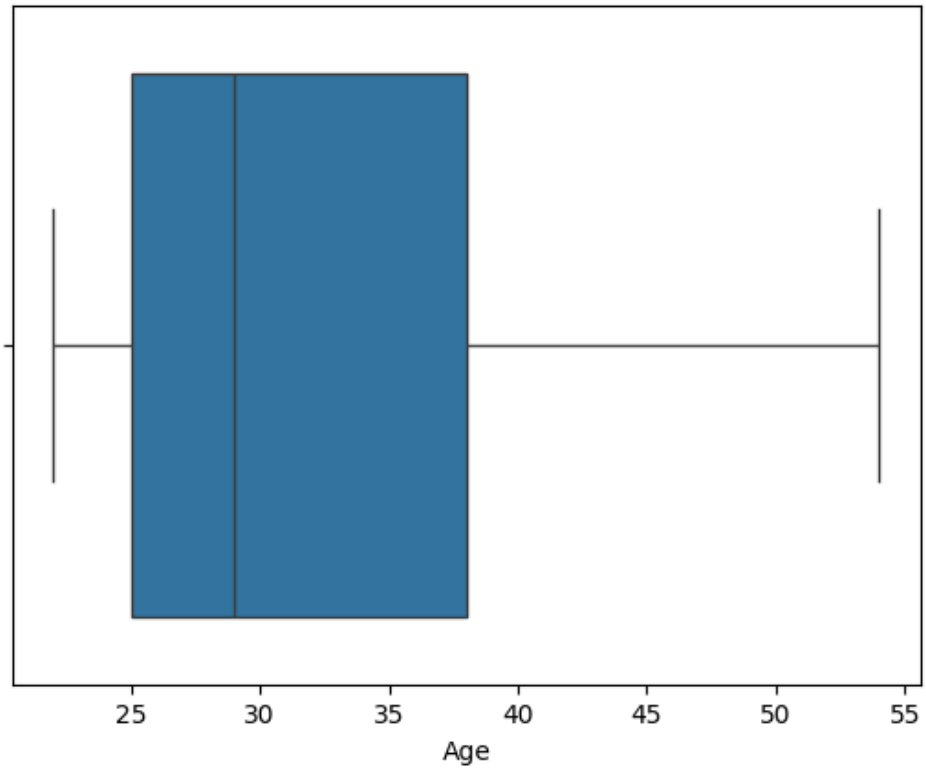


Fig16&17. Source: Austo Automobiles Notebook

Age is right-skewed: more than half of customers are between 25 – 38 years old, with a modal cluster at 23-25 years. A smaller, secondary cluster appears around 40 years, and the distribution tapers off by 54 years. No age outliers are present, indicating clean data. Marketing strategies should primarily target the 25-35 demographic while keeping an eye on the mid-career niche (38-45)

6. Price

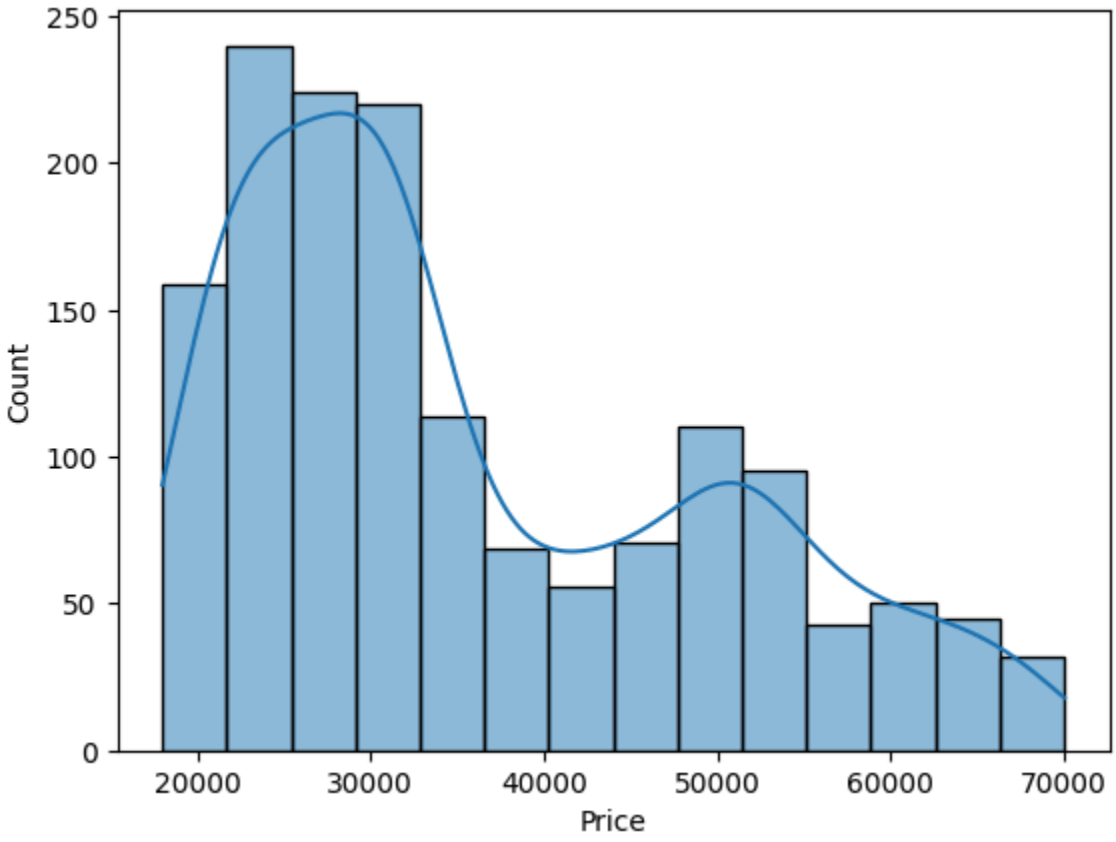


Fig18. Source: Austo Automobiles Notebook

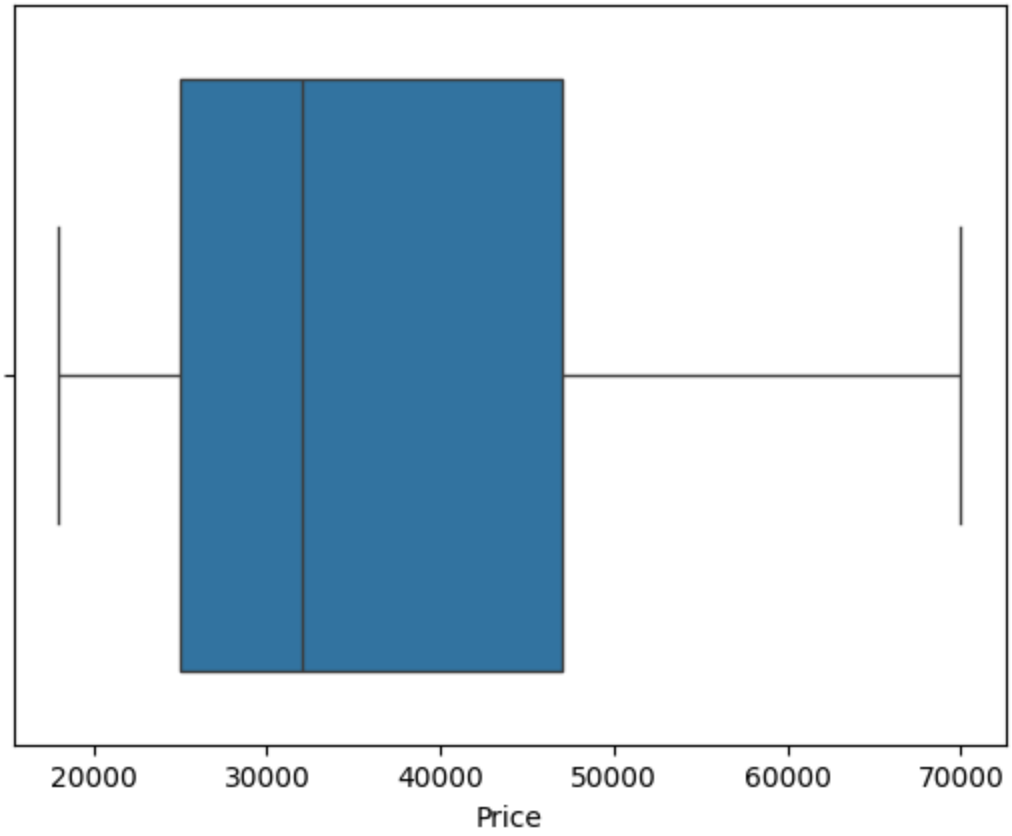
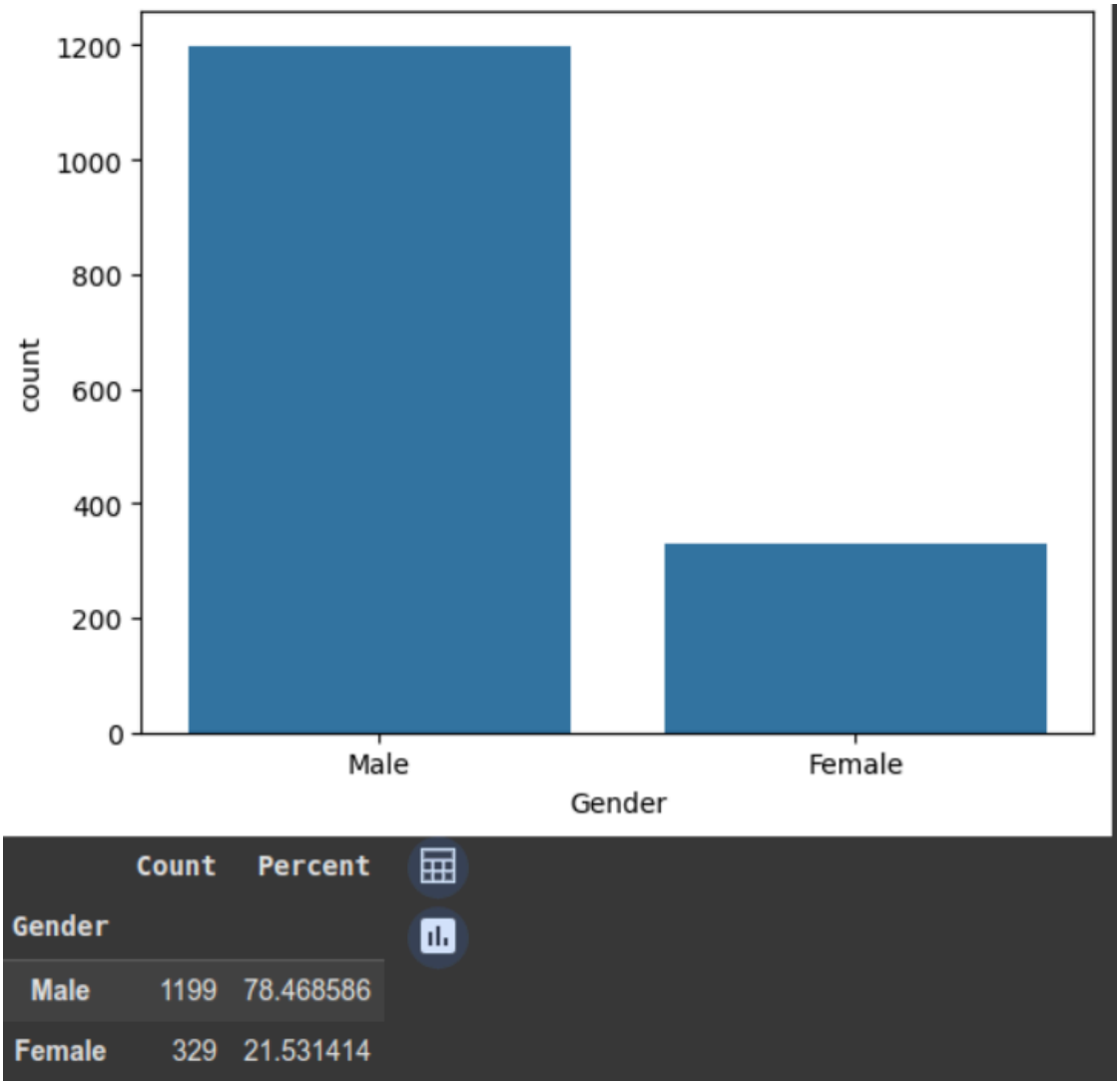


Fig19. Source: Austo Automobiles Notebook

Vehicle Price is right-skewed, with a primary concentration between ₹25 k–35 k and a smaller premium cluster around ₹50 k. The median purchase price is ₹32 k and 75 % of purchases cost <= ₹47 k. No statistical outliers appear, indicating clean, plausible pricing data. The distribution suggests two clear customer segments budget and mid-premium which can guide differentiated marketing or financing offers.

7. Gender

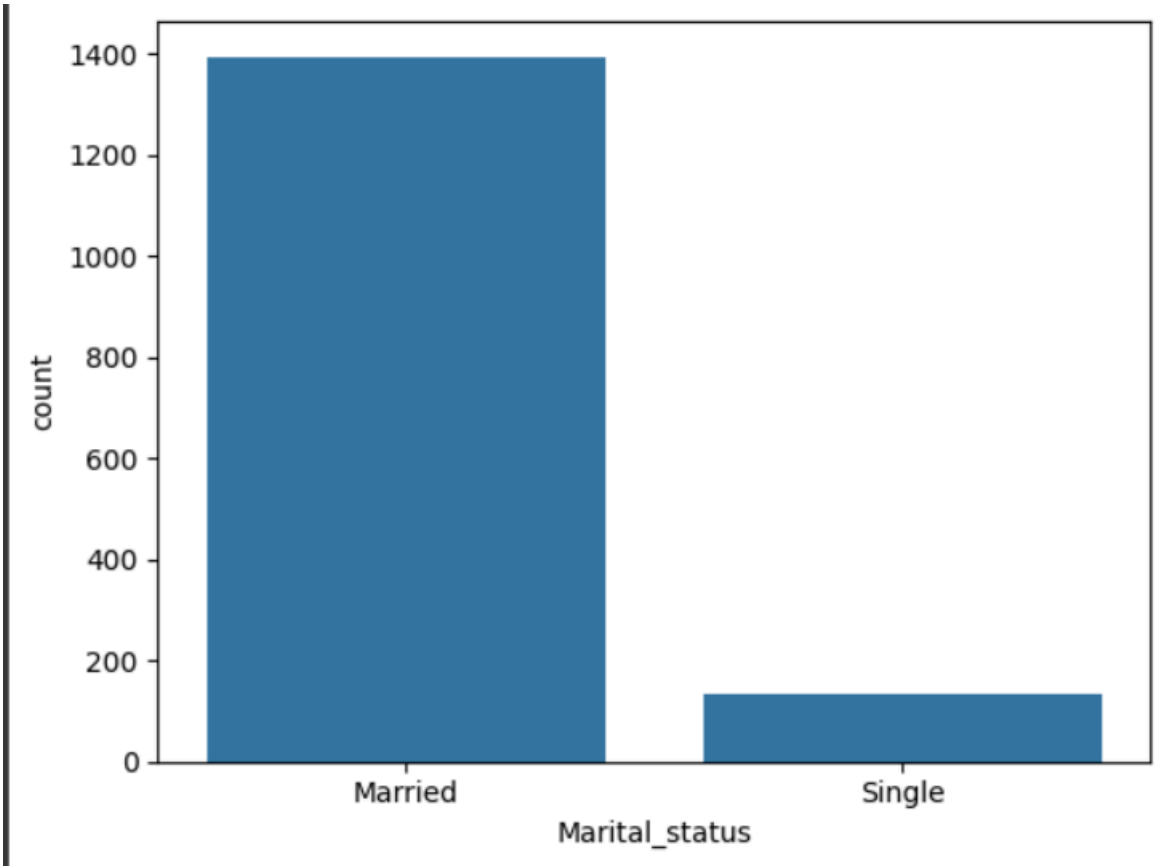


Gender	Count	Percent
Male	1199	78.468586
Female	329	21.531414

Fig20. Source: Austo Automobiles Notebook

The dataset is 78 % male and 22 % female, highlighting a substantial gender imbalance. All subsequent analyses involving gender will need to account for this skew. This disproportion also frames our first business question whether men truly show a stronger preference for SUVs than women by providing the base rates for comparison.

8. Marital_status

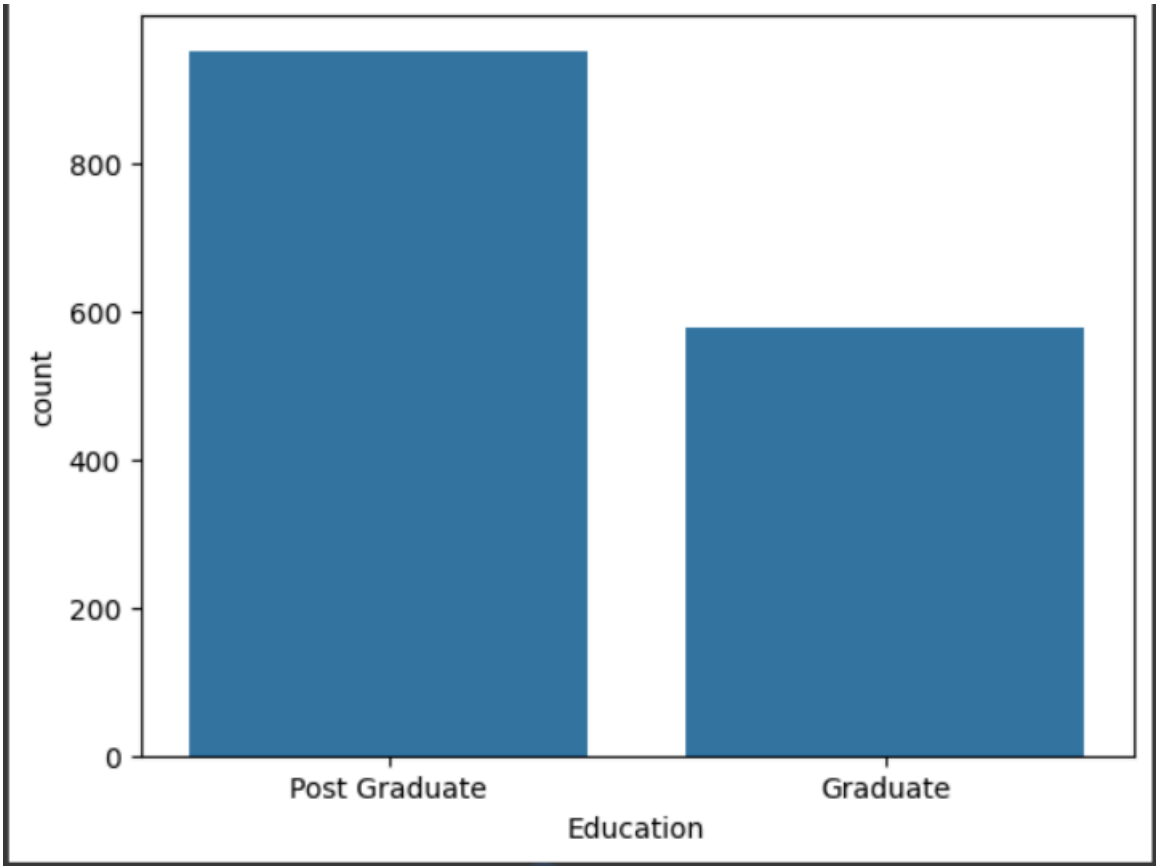


Marital_status	Count	Percent
Married	1395	91.295812
Single	133	8.704188

Fig21. Source: Austo Automobiles Notebook

The dataset is overwhelmingly composed of married individuals (91.3%), with only 8.7% reporting as single. This aligns with prior findings such as the high number of dependents and the presence of working partners.

9. Education

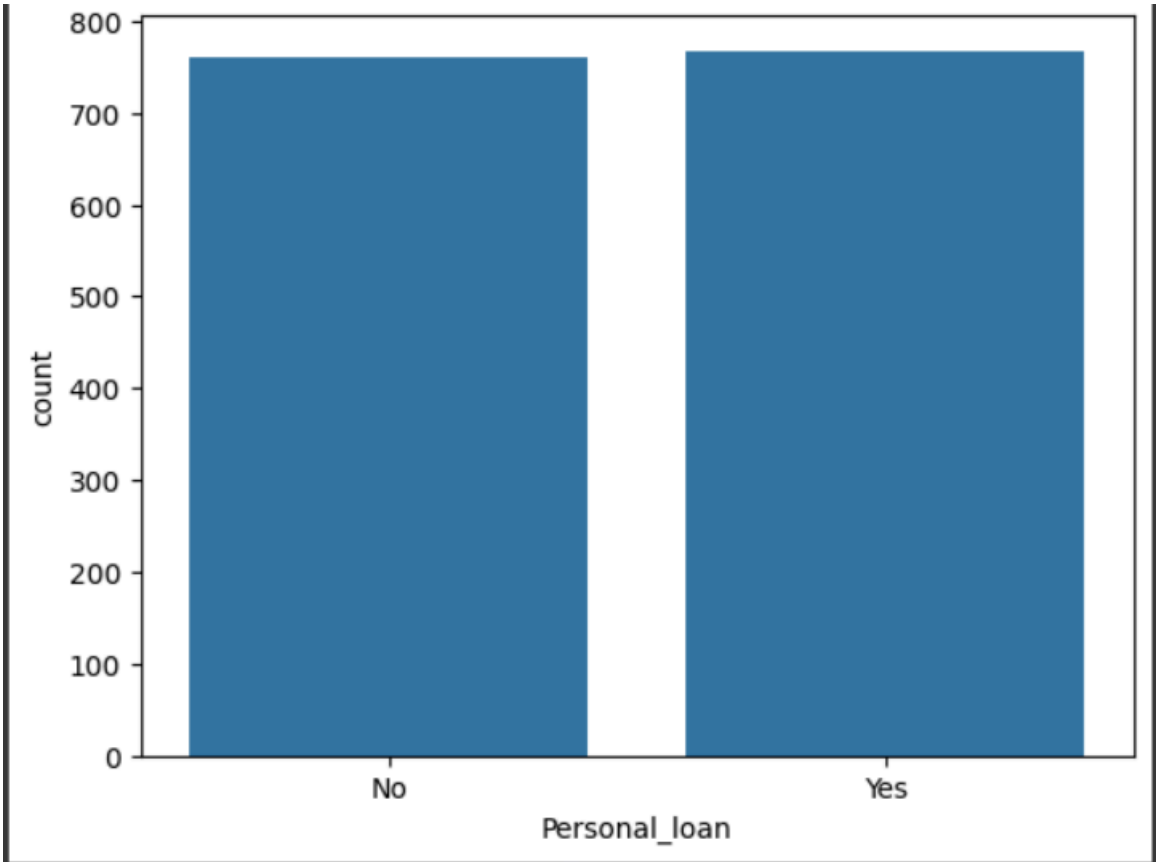


Education	Count	Percent
Post Graduate	951	62.23822
Graduate	577	37.76178

Fig22. Source: Austo Automobiles Notebook

A majority of customers (62.2%) hold postgraduate degrees, while the remaining 37.8% have undergraduate qualifications. The dataset represents a well educated audience, likely to value advanced product features, financing plans, and long-term brand value. The education variable is balanced enough to allow subgroup comparisons in spending or loan preferences.

10. Personal_loan

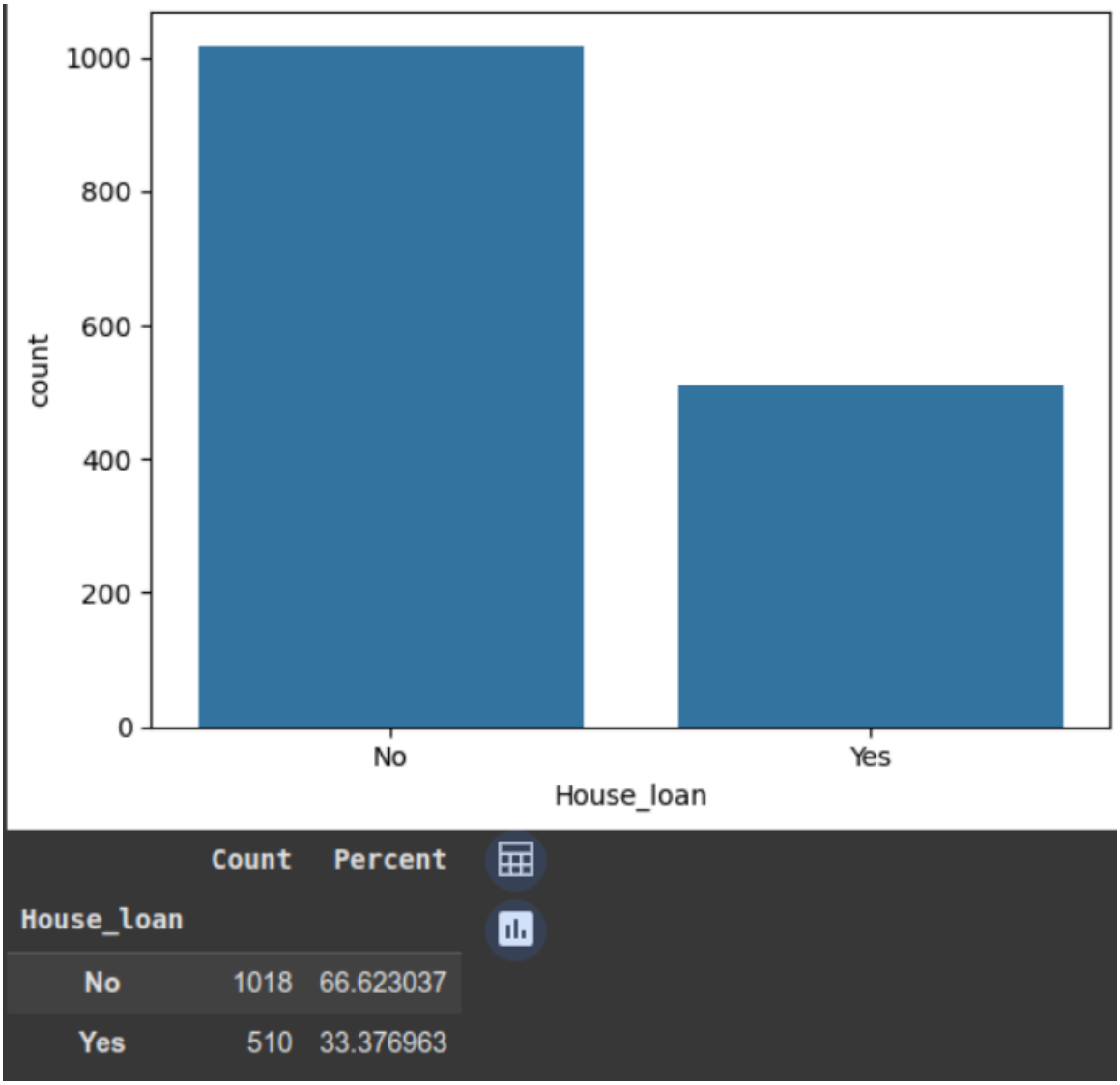


Personal_loan	Count	Percent
Yes	768	50.26178
No	760	49.73822

Fig23. Source: Austo Automobiles Notebook

Half the customers in the dataset (50.3%) have taken a personal loan, indicating strong usage of financing mechanisms in vehicle purchases. The near-even distribution makes this a valuable categorical variable for comparative analysis across other dimensions such as salary, car price, or education level.

11. House_loan

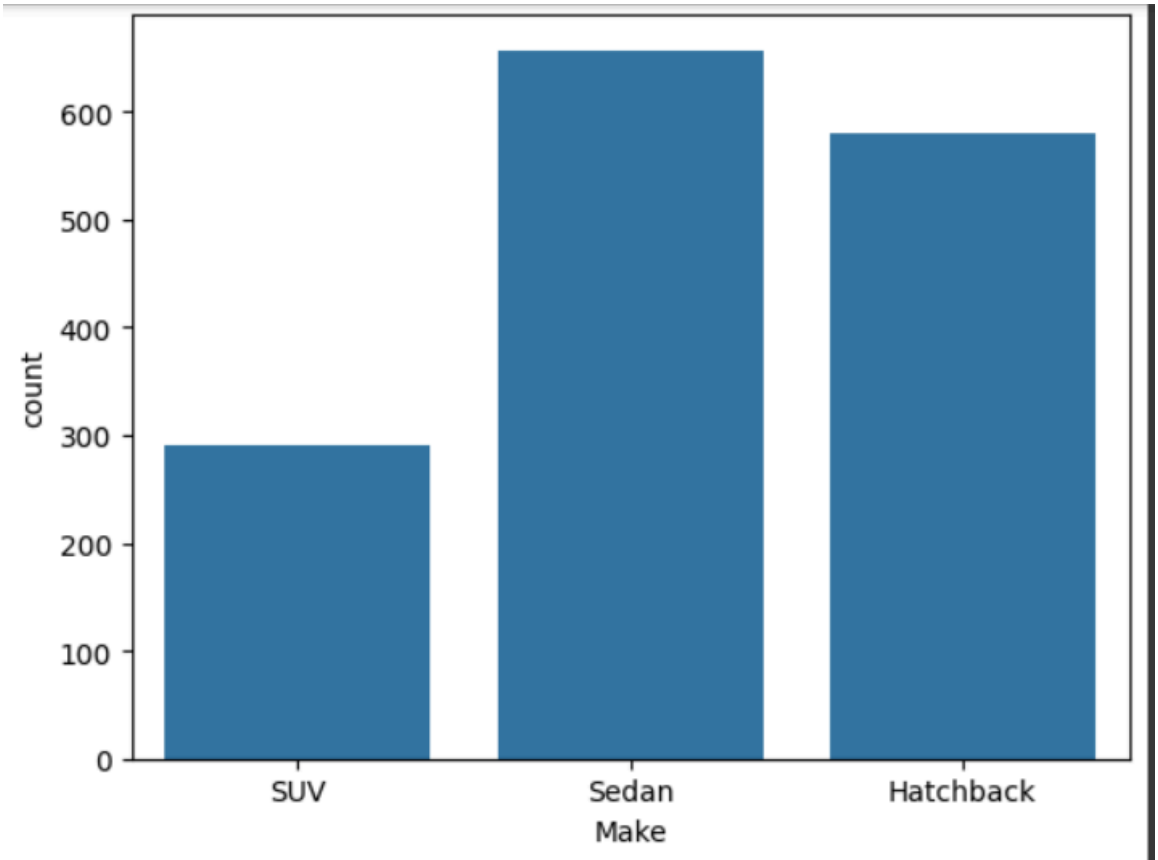


House_loan	Count	Percent
No	1018	66.623037
Yes	510	33.376963

Fig24. Source: Austo Automobiles Notebook

Roughly one-third (33.4%) of the customers have an existing house loan, while two-thirds do not. This distinction can be valuable when exploring risk segmentation, purchasing power, or cross-loan behavior. Customers without a house loan may have more disposable income for automobile spending, while those with home loans might rely more on financing

12. Make



Make	Count	Percent
Sedan	657	42.997382
Hatchback	580	37.958115
SUV	291	19.044503

Fig25. Source: Austo Automobiles Notebook

The most common vehicle type purchased is Sedan (43%), followed by Hatchback (38%). SUVs make up the smallest share (19%), indicating they are a less popular choice among the overall customer base. This makes the Make variable an important target for subgroup analysis — particularly across Gender and Profession — to see whether SUV preference varies by demographic.

Bivariate Analysis

1. **Heatmap exploring correlation among all numerical variables**



Fig26. Source: Austo Automobiles Notebook

We dropped "Total_salary" from the heatmap to avoid multicollinearity, as it is a derived variable created by summing Salary and Partner_salary. Keeping both base variables allows for more granular analysis.

Observations & Insights:

- Age vs Price

Correlation coefficient: +0.80

Insight: Strongest positive correlation in the dataset. Older customers tend to buy significantly more expensive vehicles. Likely linked to financial maturity and purchasing power.

- Salary vs Price

Correlation coefficient: +0.41

Insight: Moderate correlation; higher salary is associated with higher car prices, though other variables (e.g., age, loan status) also contribute.

- Partner Salary vs Price

Correlation coefficient: +0.17

Insight: Weak correlation. A working partner has minimal direct impact on the price of the car purchased by the individual.

- Salary vs Age

Correlation coefficient: +0.62

Insight: Expected trend — older individuals typically earn more, likely due to career progression.

- No. of Dependents vs Other Variables

Very weak or negligible correlations ($|r| < 0.20$)

Insight: Number of dependents does not significantly influence salary, partner income, or car price. This variable may be more relevant for descriptive analysis than prediction.

Conclusion: The heatmap confirms no multicollinearity, as no pair of features exceeds a threshold of |0.8|, except for Age vs Price which is a meaningful, actionable insight rather than a redundancy.

2. **Pairplot exploring relationship among all numerical variables**

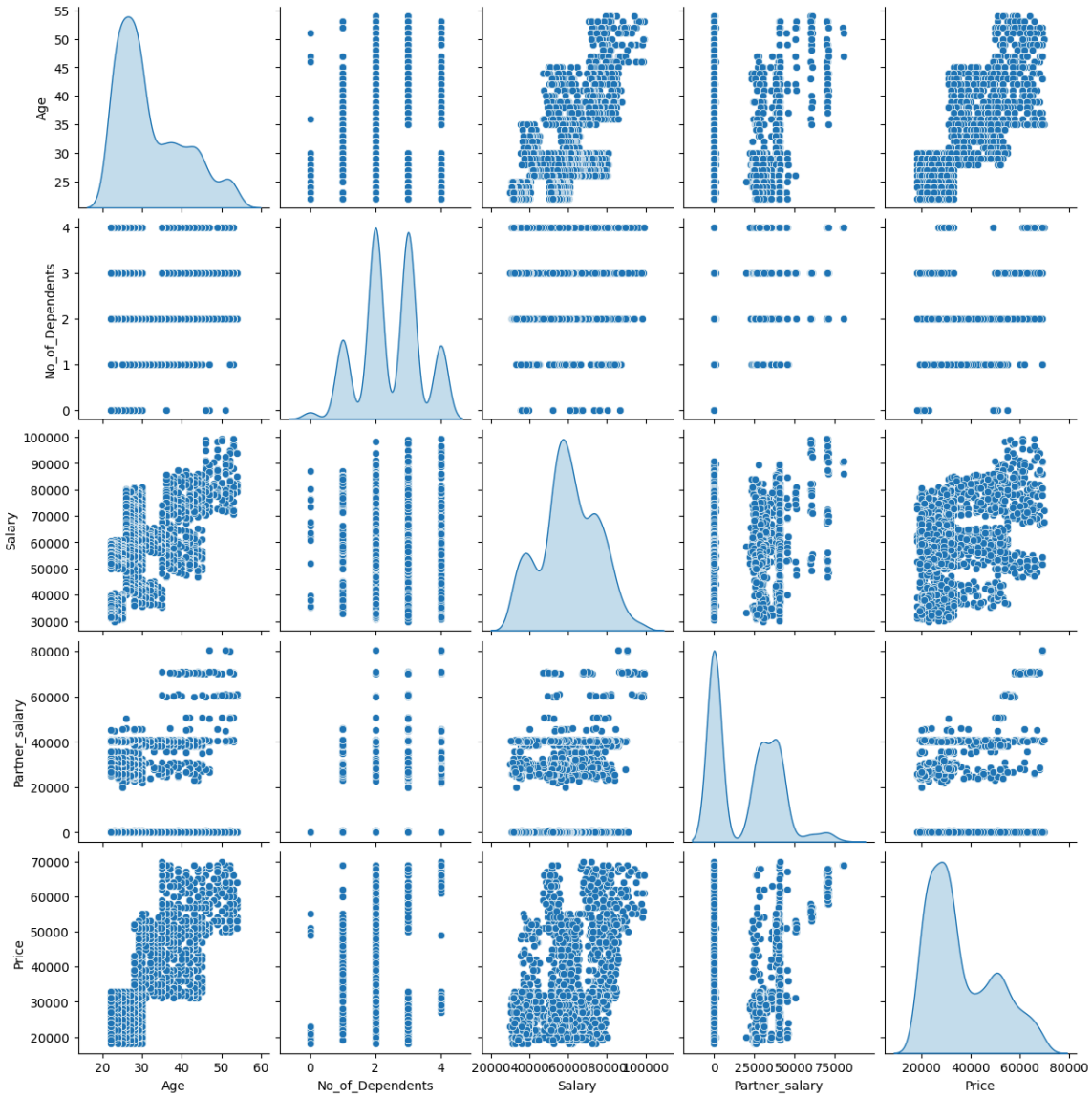
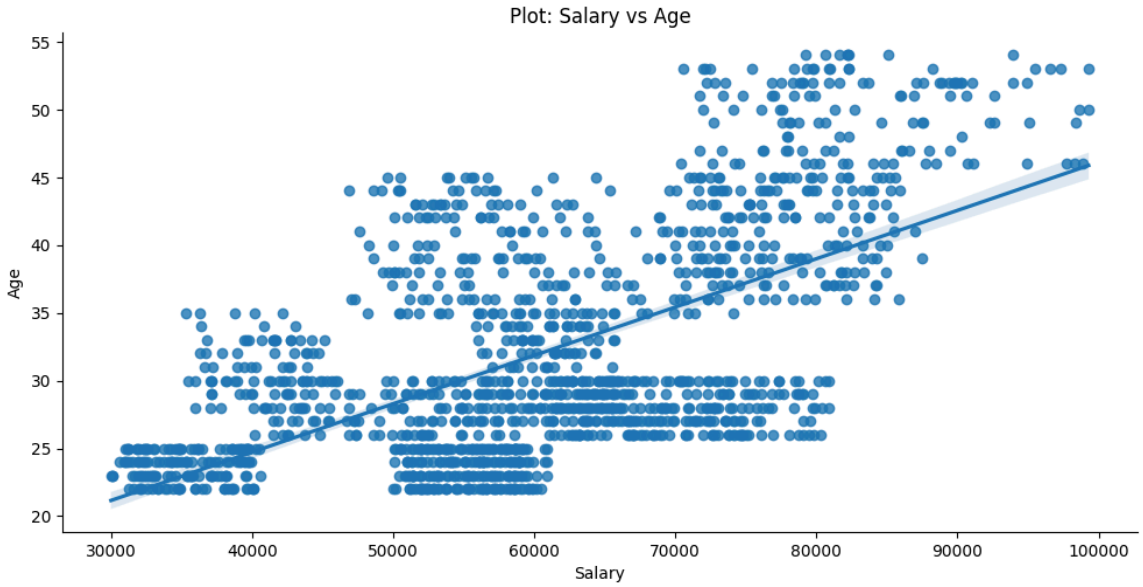


Fig27. Source: Austo Automobiles Notebook

The pairplot visually confirms the correlations observed in the heatmap, particularly the strong positive relationships between Age & Salary and Salary & Price.

3. Bivariate Analysis between strongest correlated variables (Age vs Price, Salary vs Age, Salary vs Price)





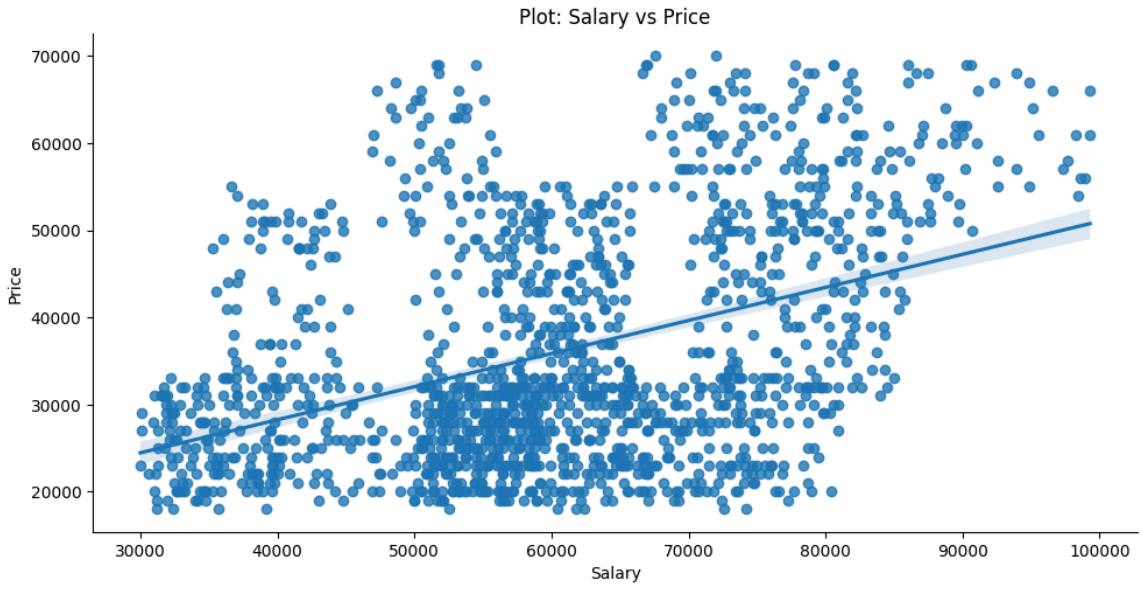


Fig 28,29&30. Source: Austo Automobiles Notebook

Analysis:

Fig 28 – Age vs Price

Clear positive linear trend: vehicle price rises steadily with age.

Slope ≈ 800 ₹ per year (visual estimate).

Practical insight Older buyers represent the high-ticket segment; premium models or upsell packages should target 40 + customers.

Fig 29 – Salary vs Age

Moderate upward trend consistent with career progression.

Spread widens after ~₹70 k — senior roles have more salary dispersion.

Fig 30 – Salary vs Price

Positive but weaker relationship than Age-Price: many mid-salary buyers still choose inexpensive cars.

Clusters — two loose bands are visible:

₹30 k–45 k salary buyers concentrated below ₹40 k price.

₹60 k + salary buyers spread widely up to ₹70 k price.

Business takeaway: Salary alone is not a strong price determinant; financing options and life-stage (age) matter more.

Overall Insights

Age is a stronger spending predictor than salary. Older customers consistently purchase higher-priced vehicles, even when salaries overlap with younger cohorts.

Salary explains career progression but not all buying power. Many mid-salary individuals still opt for lower-priced models, suggesting conservatism or loan aversion.

These focused plots corroborate the heat-map findings and will inform our feature selection and segmentation strategy in the subsequent analysis steps.

Fig 29 – Age vs Price

Clear positive linear trend: vehicle price rises steadily with age.

Slope ≈ 800 ₹ per year (visual estimate).

Practical insight Older buyers represent the high-ticket segment; premium models or upsell packages should target 40 + customers.

Fig 29 – Salary vs Age

Moderate upward trend consistent with career progression.

Spread widens after ~₹70 k — senior roles have more salary dispersion.

Fig 30 – Salary vs Price

Positive but weaker relationship than Age-Price: many mid-salary buyers still choose inexpensive cars.

Clusters — two loose bands are visible:

₹30 k–45 k salary buyers concentrated below ₹40 k price.

₹60 k + salary buyers spread widely up to ₹70 k price.

Business takeaway: Salary alone is not a strong price determinant; financing options and life-stage (age) matter more.

Overall Insights

Age is a stronger spending predictor than salary. Older customers consistently purchase higher-priced vehicles, even when salaries overlap with younger cohorts.

Salary explains career progression but not all buying power. Many mid-salary individuals still opt for lower-priced models, suggesting conservatism or loan aversion.

These focused plots corroborate the heat-map findings and will inform our feature selection and segmentation strategy in the subsequent analysis steps.

4. **Categorical values vs Numerical values bivariate analysisCategorical values vs Numerical values bivariate analysis**

- *Salary Distribution by Gender*

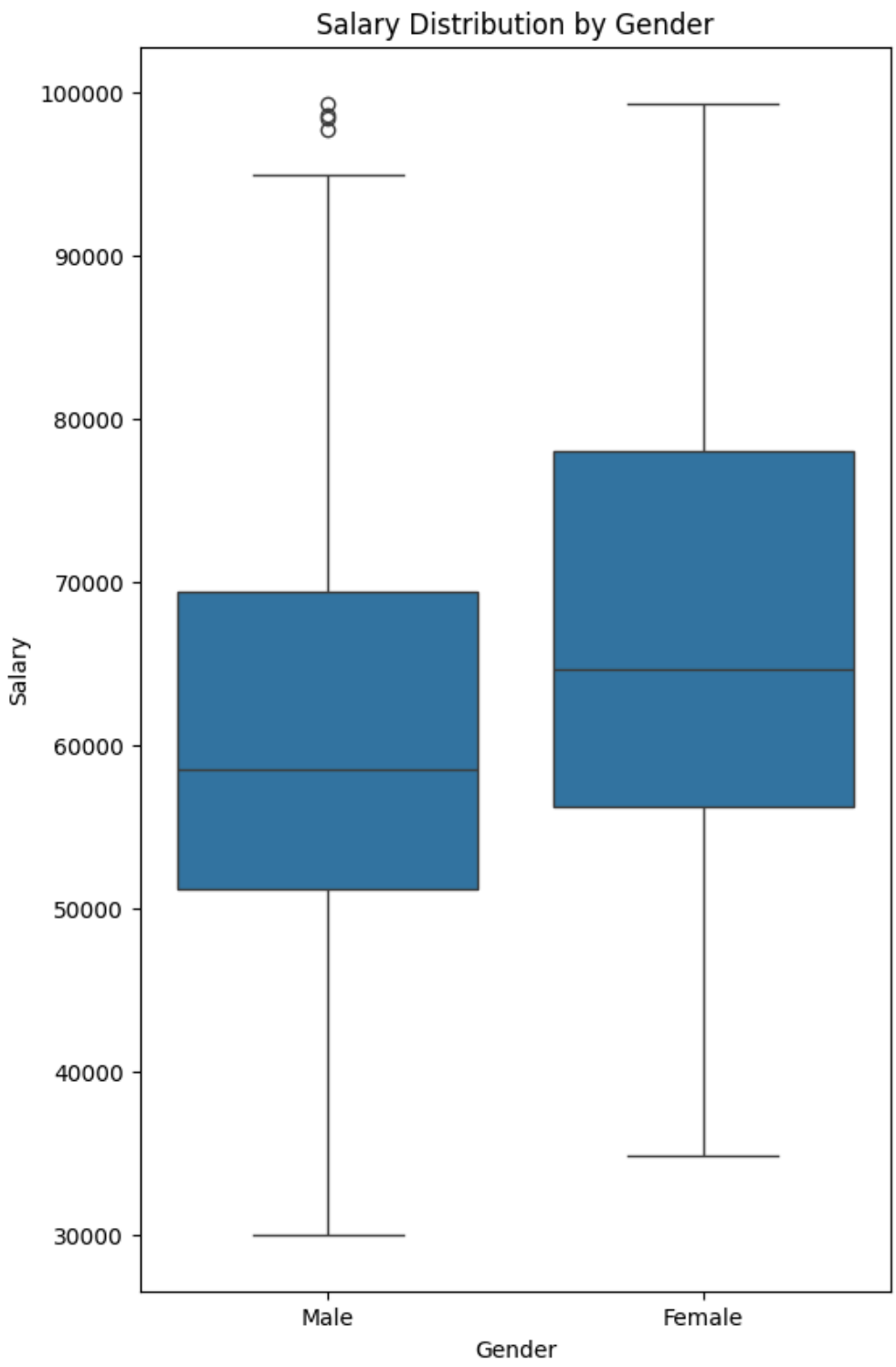


Fig31. Source: Austo Automobiles Notebook

- *Salary distribution by Profession*

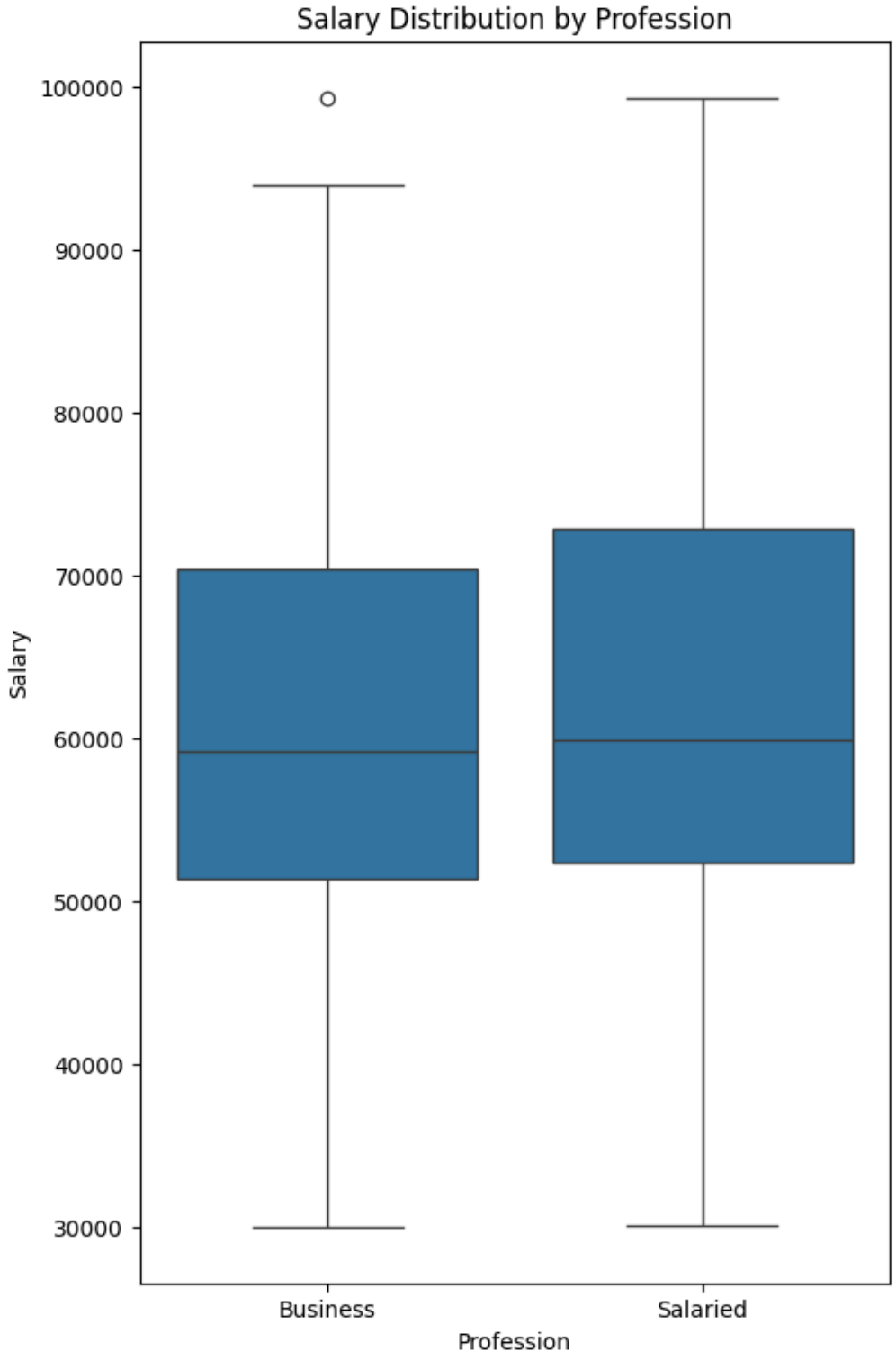


Fig32. Source: Austo Automobiles Notebook

- *Salary Distribution by Personal Loan*

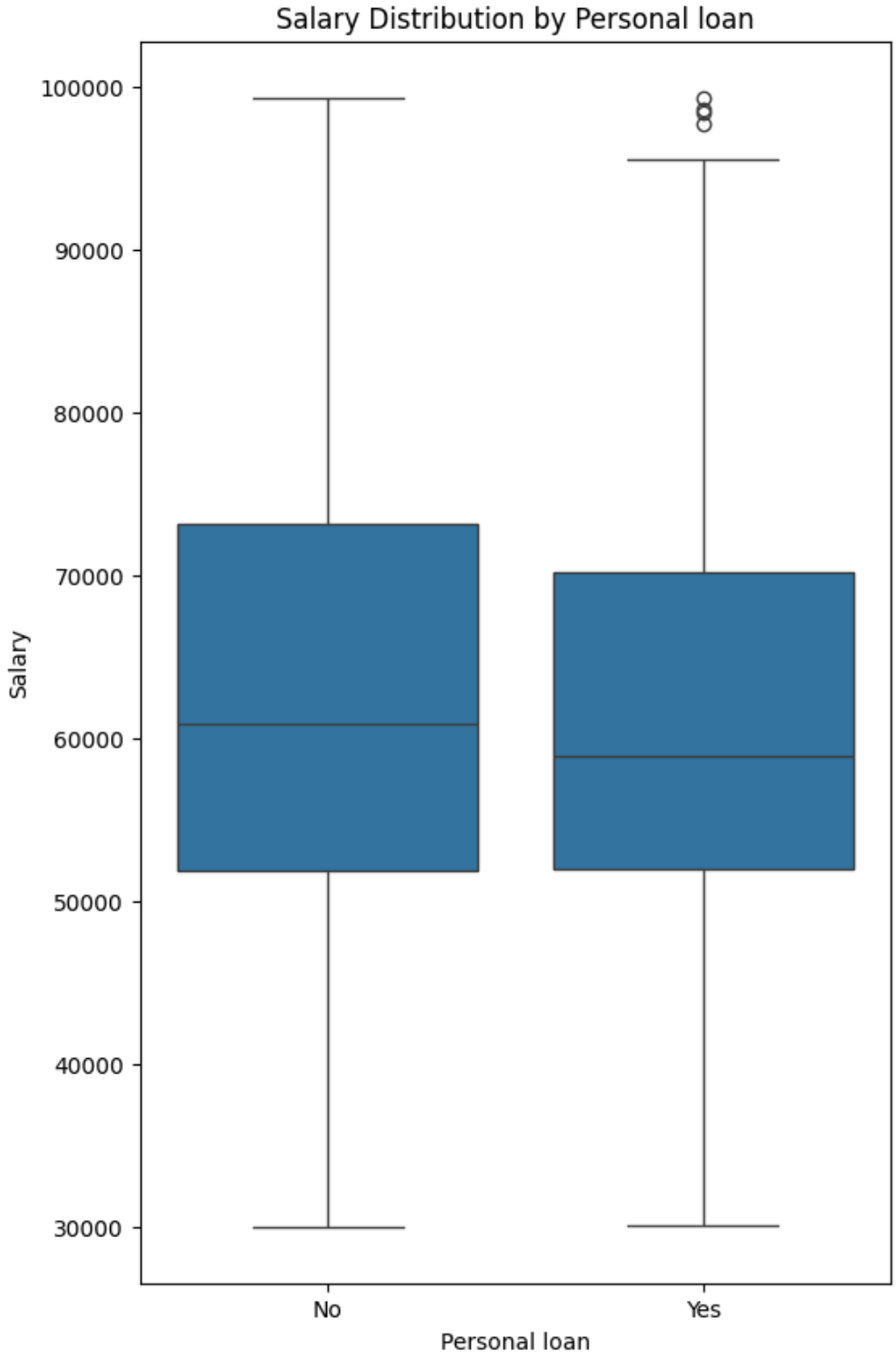


Fig33. Source: Austo Automobiles Notebook

- *Salary Distribution by Education*

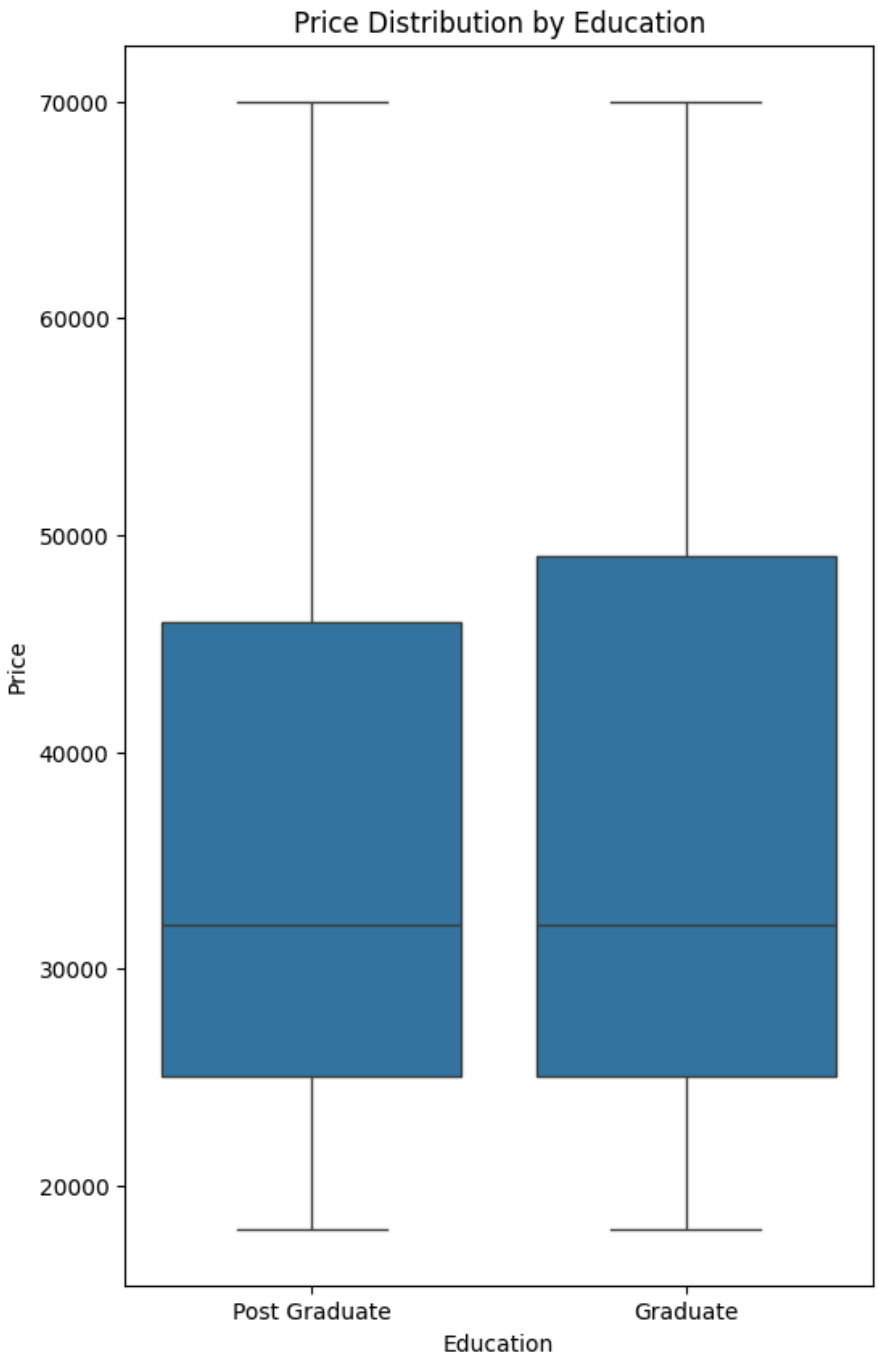


Fig34. Source: Austo Automobiles Notebook

- *Price distribution by Make*

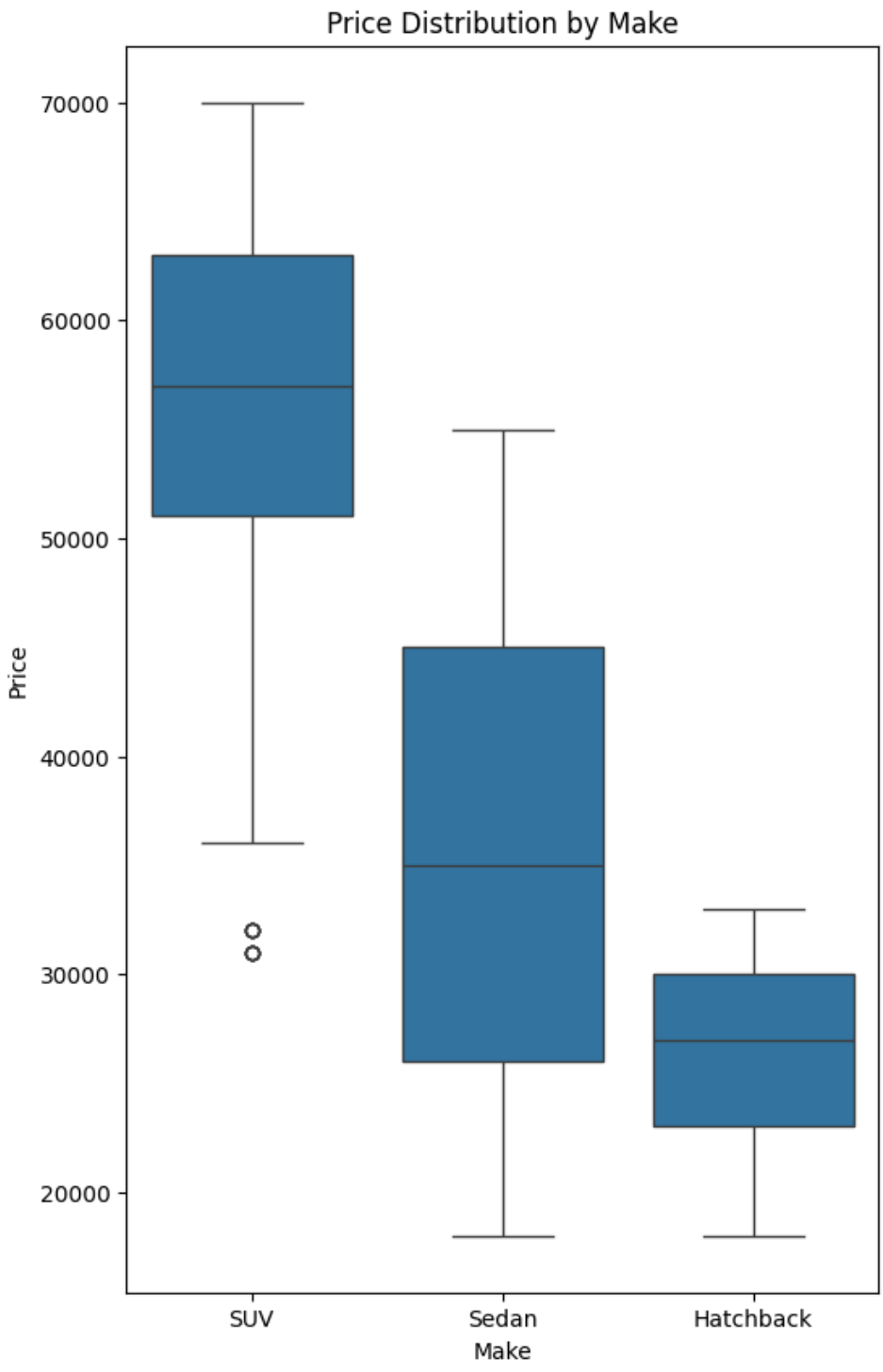


Fig35. Source: Austo Automobiles Notebook

Observations:

Price vs Make

Box plot reveals that:

SUVs have the highest median price.

Sedans are priced mid-range.

Hatchbacks are generally low-cost.

This suggests a clear segmentation of the market by car type.

Price vs Education

Median price is nearly the same for both Graduate and Post Graduate groups.

This implies education level has minimal impact on the amount spent on cars.

Salary vs Personal Loan

Median salaries for customers with and without personal loans are similar.

This indicates personal loan decisions are not driven by income, but potentially by spending habits or other factors.

Salary vs Profession

Salaried and Business professionals show similar salary distributions.

High-income outliers exist in both categories.

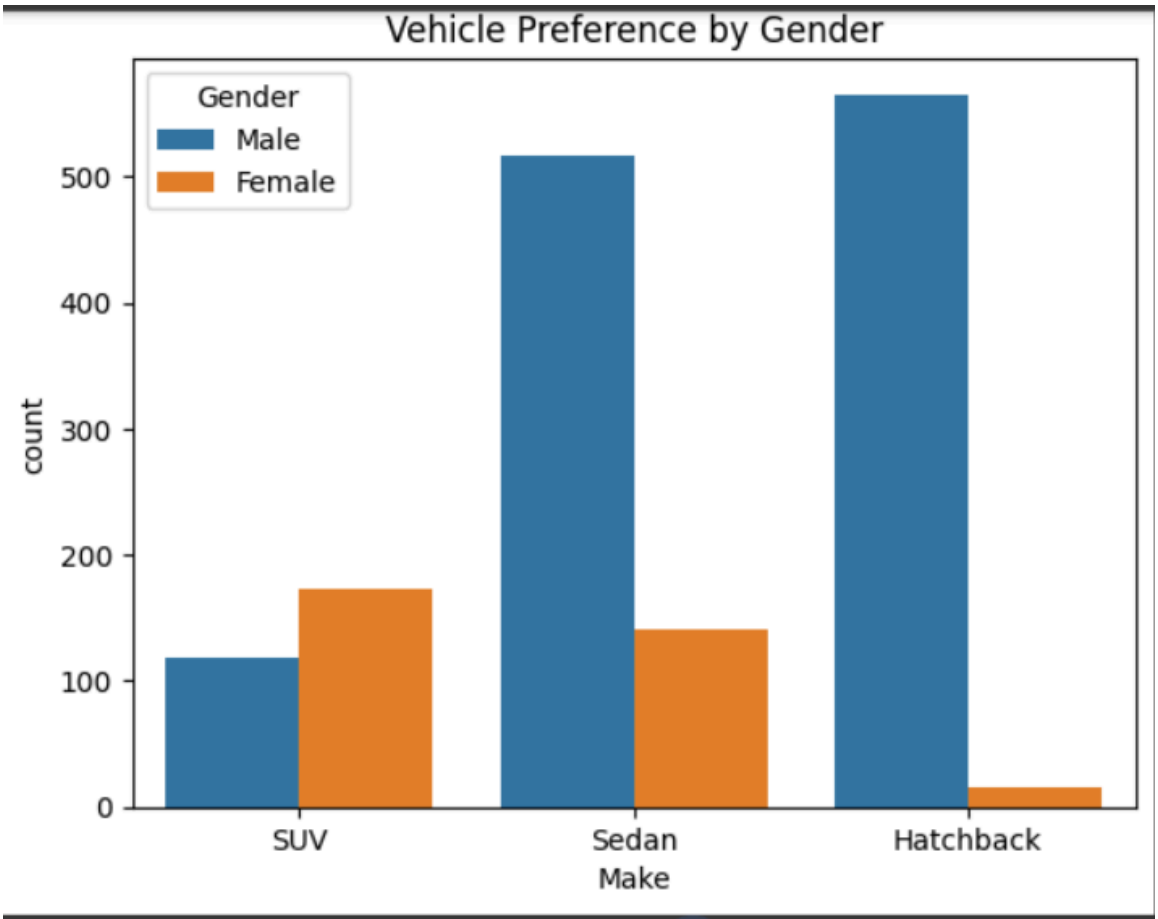
Salary vs Gender

Median salary of females is slightly higher than that of males.

Gender does not create a major divide in income in this sample, suggesting marketing and financing offers should be gender-neutral.

5. **Exploring the data to answer key questions using bivariate analysis**

Q1. Do men tend to prefer SUVs more compared to women?



Make	Hatchback	SUV	Sedan
Gender			
Female	4.559271	52.583587	42.857143
Male	47.122602	9.841535	43.035863

Fig36. Source: Austo Automobiles Notebook

To answer this, we performed a bivariate analysis between the categorical variables: Make (vehicle type) and Gender.

We used a countplot to visualize the distribution of vehicle preferences by gender.

We created a normalized crosstab to compute the percentage distribution of each vehicle type within each gender.

A majority of females (52.6%) prefer SUVs, while only 9.8% of males prefer them.

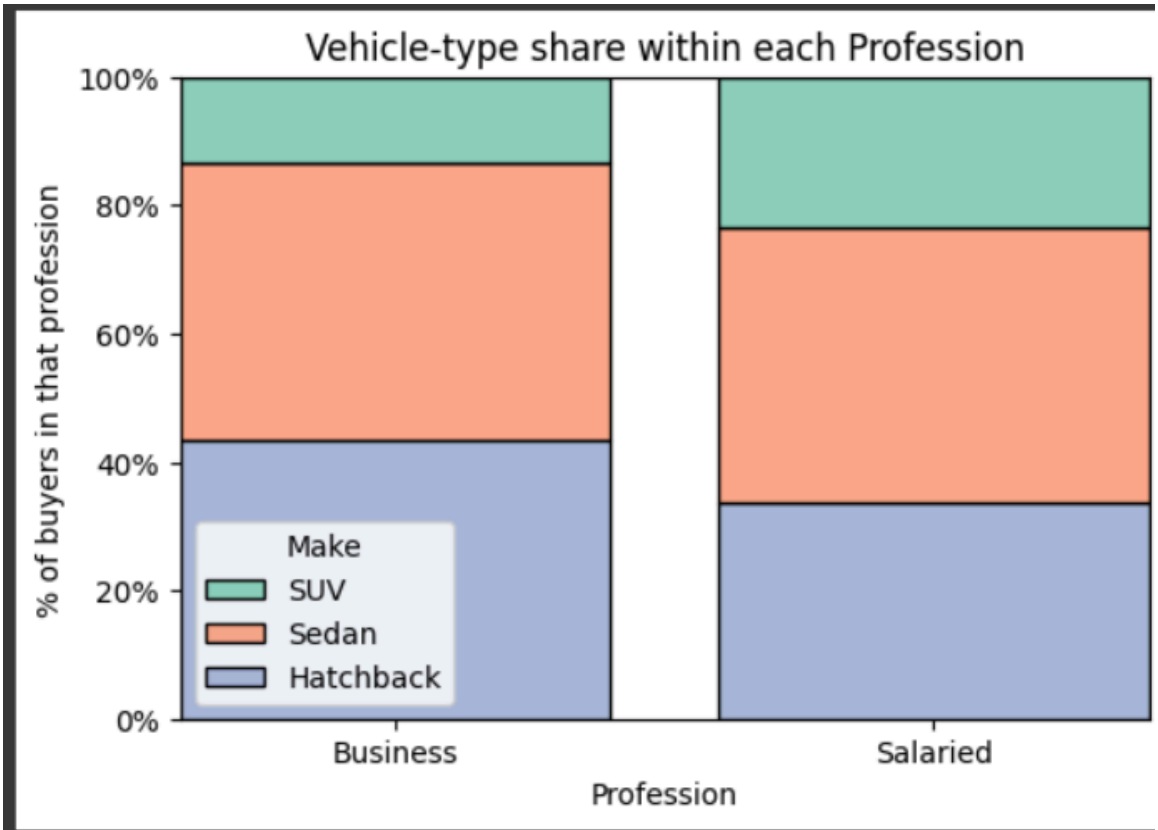
On the contrary, males heavily prefer Hatchbacks (47.1%), whereas only 4.6% of females prefer them.

Sedans are roughly equally preferred by both genders (~43%).

Conclusion: No, men do not tend to prefer SUVs more than women.

In fact, the data shows that SUVs are significantly more popular among women, while men tend to prefer Hatchbacks instead.

Q2. What is the likelihood of a salaried person buying a Sedan?




```
Make
Sedan        42.824074
Hatchback    33.680556
SUV          23.495370
Name: proportion, dtype: float64
```

Fig37. Source: Austo Automobiles Notebook

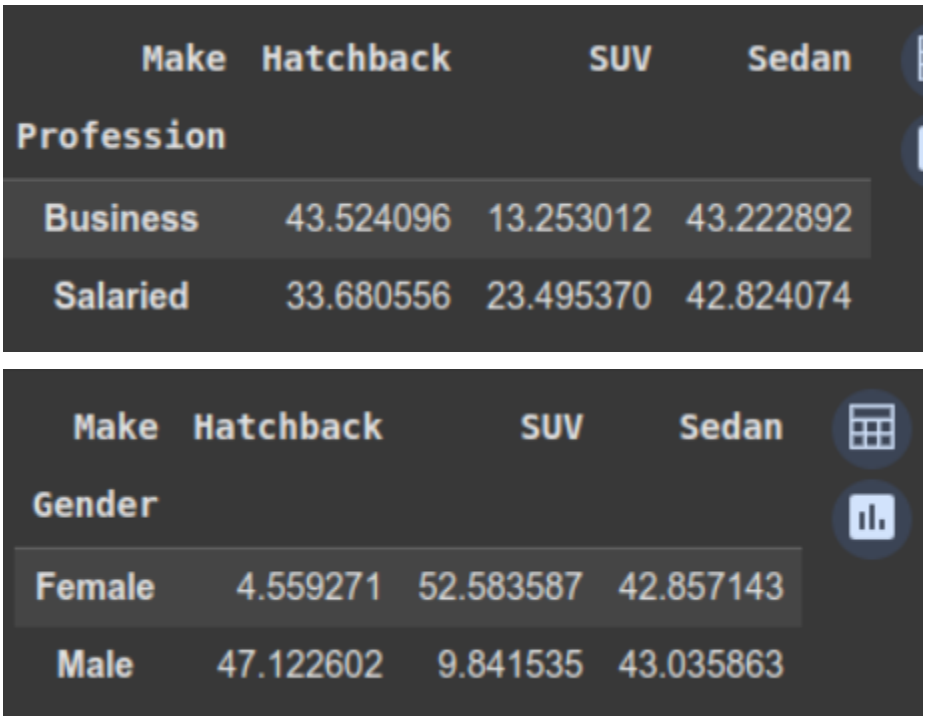
≈ 43 % of salaried buyers choose a Sedan.

Hatchbacks trail at ~34 %, SUVs at ~23 %.

Therefore, the likelihood (probability) that a randomly selected salaried customer in this dataset buys a Sedan is about 1 in 2.3.

Business take-away: Sedans are the dominant choice among salaried professionals; marketing bundles or financing plans aimed at this segment should prioritise Sedan models.

Q3. What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?



Make	Hatchback	SUV	Sedan
Profession			
Business	43.524096	13.253012	43.222892
Salaried	33.680556	23.495370	42.824074

Make	Hatchback	SUV	Sedan
Gender			
Female	4.559271	52.583587	42.857143
Male	47.122602	9.841535	43.035863

Fig38&39. Source: Austo Automobiles Notebook

Based on our previous bivariate analysis, Sheldon Cooper's claim is not supported by the data. Let’s look at the evidence:

Females overwhelmingly prefer SUVs (52.6%). Males only show 9.8% SUV preference; they heavily prefer Hatchbacks and Sedans.

Even among salaried individuals, the SUV share is lowest (23.5%), with Sedan being the top choice.

Conclusion: No evidence supports the idea that a salaried male is an easier SUV sale. In fact, both the gender and profession trends indicate that:

Males lean toward Hatchbacks and Sedans

SUVs appeal more to females than malesSalaried individuals show higher preference for Sedans. Thus, Sheldon Cooper is wrong — at least according to this dataset.

Q4. How does the the amount spent on purchasing automobiles vary by gender?

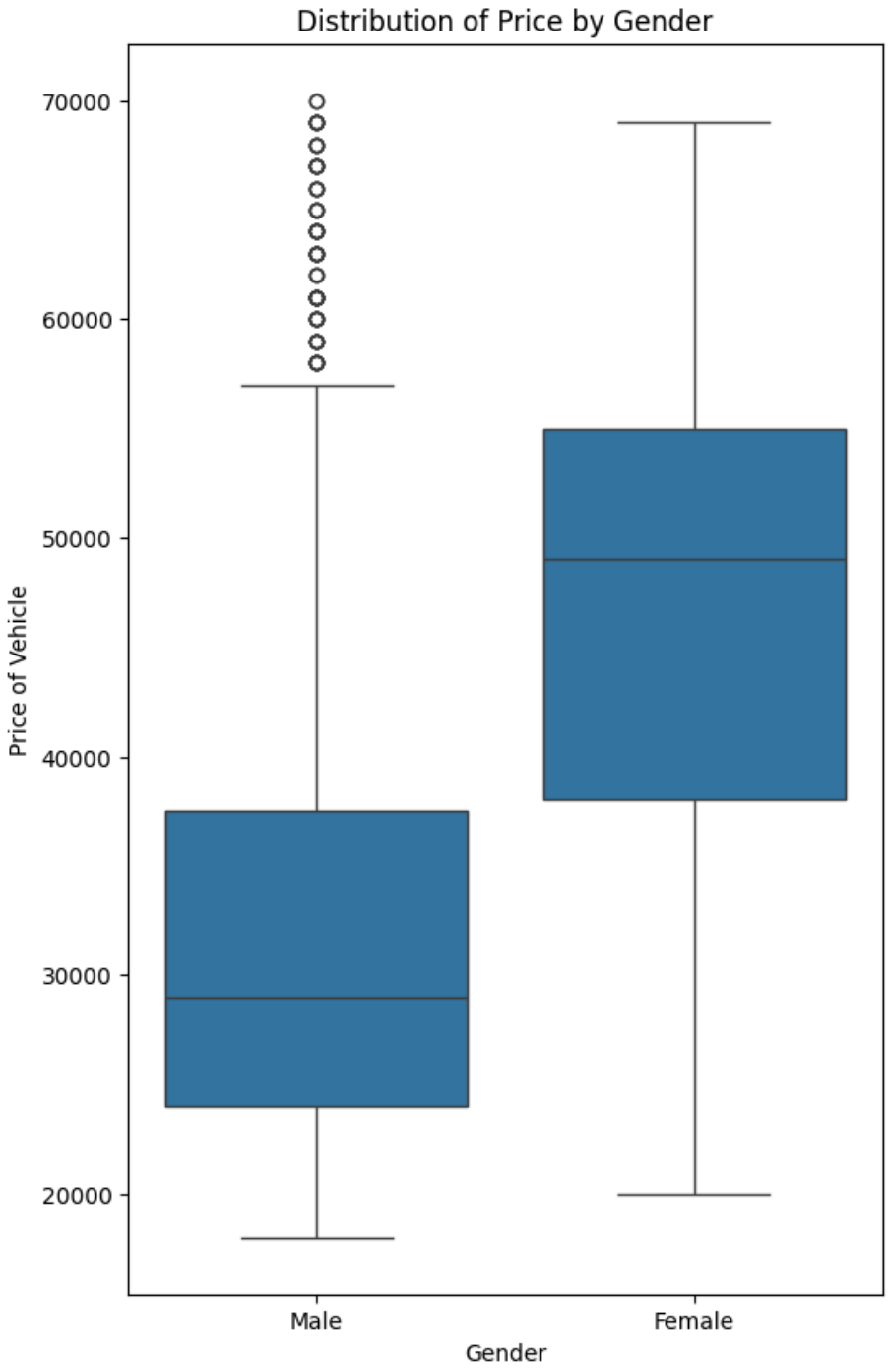
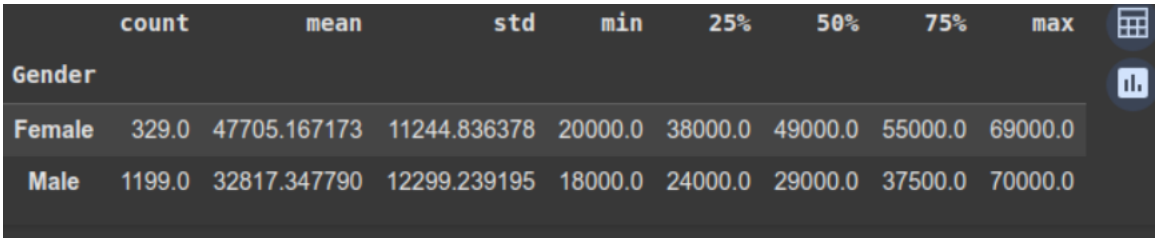


Fig40. Source: Austo Automobiles Notebook



Gender	count	mean	std	min	25%	50%	75%	max
Female	329.0	47705.167173	11244.836378	20000.0	38000.0	49000.0	55000.0	69000.0
Male	1199.0	32817.347790	12299.239195	18000.0	24000.0	29000.0	37500.0	70000.0

Fig41. Source: Austo Automobiles Notebook

Females spend markedly more.

Median purchase price for women is ₹49 k, vs ₹29 k for men.

Mean price shows the same gap (≈ ₹47.7 k vs ₹32.8 k).

Distribution shift, not just outliers.

The entire IQR for females (38 k – 55 k) sits above the male IQR (24 k – 37.5 k).

Both groups share similar upper limits (~ ₹70 k), so females are not skewed by a few luxury buyers—they systematically buy dearer vehicles.

Sample-size note.

Males outnumber females 3.6 : 1, yet still spend less per vehicle on average.

Conclusion: Women in this dataset spend roughly ₹17 k more per vehicle than men (median basis). They form a smaller segment but represent higher-revenue transactions. Marketing higher-trim models, premium features and extended-warranty bundles toward female buyers could lift overall revenue.

Q5. How much money was spent on purchasing automobiles by individuals who took a personal loan?



Fig42. Source: Austo Automobiles Notebook

Personal_loan	Total_Spend	Avg_Spend	Count
No	28399000	37367.105263	760
Yes	26644000	34692.708333	768

Total amount spent by personal-loan customers: ₹26,644,000

Fig43. Source: Austo Automobiles Notebook

Total outlay: Personal-loan customers spent ₹ 26.6 million on vehicles— only ~6 % less than the non-loan group, even though their average ticket price is lower.

Average ticket size: Loan takers spend ≈ ₹ 2.7 k less per buyer than cash buyers (₹ 34.7 k vs ₹ 37.4 k).

Volume compensates: The loan group's slightly larger head-count (768 vs 760) almost offsets the lower per-car price, producing nearly comparable total revenue.

Insight: Personal-loan financing drives almost half of total sales revenue. Although each financed purchase is a bit cheaper, the sheer number of loan buyers keeps their aggregate spend nearly on par with cash buyers. Offering attractive personal-loan packages is therefore critical to sustaining volume.

Q6. How does having a working partner influence the purchase of higher-priced cars?



Fig44. Source: Austo Automobiles Notebook

Partner_working	count	mean	std	min	25%	50%	75%	max
No	690.0	36378.260870	13747.583185	18000.0	25000.0	32000.0	48000.0	70000.0
Yes	838.0	35730.310263	13366.493269	18000.0	25000.0	32000.0	46000.0	70000.0

Fig45. Source: Austo Automobiles Notebook

The average vehicle price is slightly higher for individuals whose partners are not working (₹36,378 vs ₹35,730).

However, the median price is exactly the same in both groups (₹32,000), suggesting similar central tendencies.

The maximum price is ₹70,000 in both groups.

The distribution spread is comparable as shown in the boxplot.

Visualization: A boxplot was used to compare the price distribution by partner working status.

The boxplots confirm that both groups have similar distributions with minor differences in upper quartiles.

Conclusion: Having a working partner does not significantly influence the purchase of higher-priced cars. Both groups show similar price ranges, medians, and distributions.

Strategic Recommendations

1. Target Salaried Individuals with Sedan Promotions

- A significant proportion of salaried individuals prefer Sedans.
- Introduce personalized financing options and EMI schemes for this segment to drive conversion.

2. Focus SUV Marketing on Female Customers

- The analysis reveals a higher SUV preference among women.
- Develop ad campaigns showcasing comfort, safety, and family features that appeal to female buyers.

3. Upsell Higher-Priced Models to Customers with Working Partners

- Customers with working partners tend to purchase higher-priced vehicles.
- Position premium models as aspirational upgrades for financially stable households.

4. Use Gender-Based Pricing Strategies

- Females, on average, spend more on car purchases than males.
- Introduce premium packages with enhanced features (aesthetic upgrades, safety, infotainment) tailored to female preferences.

5. Leverage Personal Loan Offers for Mid-Range Vehicles

- Those who took personal loans show a tendency to spend more on automobiles.
- Collaborate with financial institutions to provide instant loan approvals and highlight “affordable luxury” for mid-range vehicles.

6. Bundle Add-Ons for Young Buyers Purchasing Entry-Level Cars

- Younger age groups tend to buy less expensive models.
- Offer value bundles (free accessories, extended warranty, service plans) to increase perceived value without raising base price.

7. Monitor Regional Gender Ratios for Inventory Optimization

- Tailor dealership inventory by analyzing the gender distribution and purchase trends at the regional level.
- Stock more SUVs in areas with higher female customer bases.

8. Target Middle-Aged Buyers (30–45) with Premium Car Options

- The data indicates that individuals in the 30–45 age range tend to spend more on vehicles than younger or older buyers.
- Position mid-to-high-tier models (SUVs and top-end Sedans) as lifestyle upgrades for this demographic, emphasizing features like comfort, status, and long-term reliability.

Conclusion

This analysis of Austo Automobiles' customer data reveals several important insights into buyer behavior and purchasing trends. Gender, income structure, partner employment, and age all play meaningful roles in determining the type and value of vehicles purchased.

Men are more likely to purchase SUVs, whereas women tend to favor higher-priced vehicles, possibly indicating different value perceptions or lifestyle needs. Salaried individuals show a preference for Sedans, and those with working partners are inclined to spend more, potentially due to increased household income. Additionally, the age group of 30–45 emerges as a key segment for premium car purchases.

These findings offer strategic opportunities for Austo Automobiles to optimize its marketing, pricing, and product bundling efforts. By aligning promotional campaigns with these customer patterns, the company can enhance engagement, drive conversions, and boost revenue through more targeted, data-driven decisions.